

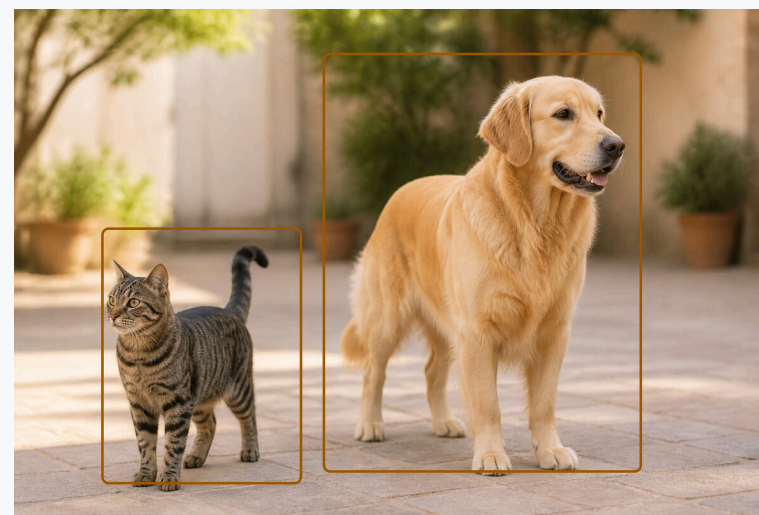
Object detection: from one label to a set of objects

VISION / OBJECT DETECTION

How can a fixed network produce a variable, unordered object set?

Prof. Nipun Batra

ES 667, IIT Gandhinagar



Why is classification not enough?

SECTION 00



What would one image-level label leave out?

One label?



cat

dog / other

One label now?



cat or dog?

Both are present.

The winning class leaves the cat out



SOFTMAX DISTRIBUTION

cat 0.46

dog 0.51

other 0.03

dog wins

Which classes are present?



INDEPENDENT SCORES

cat 0.97

dog 0.95

other 0.02

Which classes are present?



How many dogs?

Where is each one?

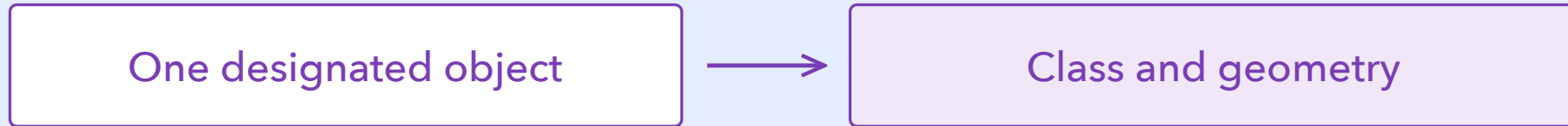
Class presence and object instances

CLASS PRESENCE

is not the same as OBJECT INSTANCES

What should one object record contain?

SECTION 01



First solve the simpler case: one designated dog.

One dog, one record



WHAT?

dog

One dog, one record



WHERE?

A box enclosing this instance.

$$\mathbf{b}_i = (x_c, y_c, w, h)$$

One dog, one record



TRAINING TARGET

$$y_i = (c_i, \mathbf{b}_i)$$

One dog, one record

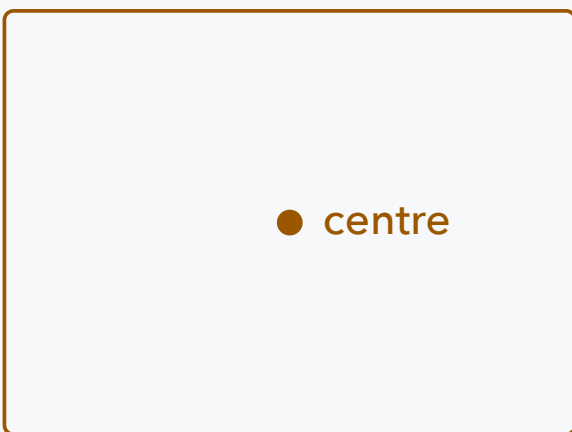


INFERENCE RECORD

$$\hat{y}_i = (\hat{c}_i, \hat{\mathbf{b}}_i, s_i)$$

The score ranks predicted records.

Centre and full box size



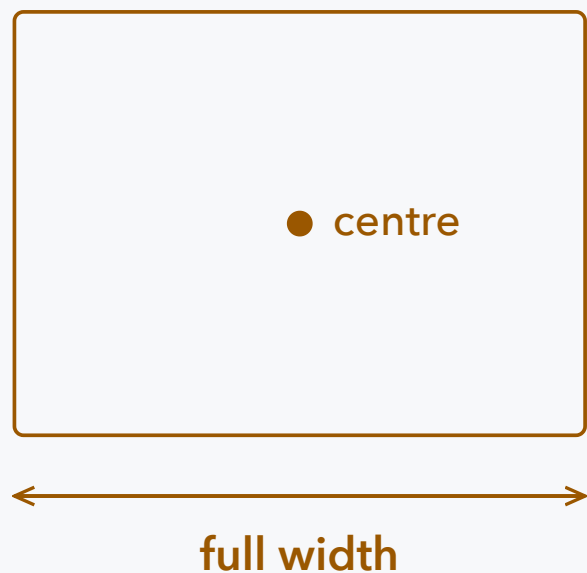
$$\mathbf{b} = (x_c, y_c, w, h)$$

Origin at the top left.

Horizontal coordinates increase rightward.

Vertical coordinates increase downward.

Centre and full box size



$$\mathbf{b} = (x_c, y_c, w, h)$$

Origin at the top left.

Horizontal coordinates increase rightward.

Vertical coordinates increase downward.

Centre and full box size



$$\mathbf{b} = (x_c, y_c, w, h)$$

Origin at the top left.

Horizontal coordinates increase rightward.

Vertical coordinates increase downward.

Normalize horizontal values by image width and vertical values by height.

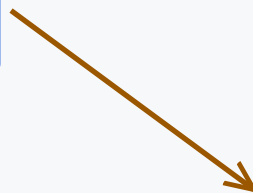
One backbone, two heads



Shared image
features



Class head: C scores



Box head: 4 values

One backbone, two heads



Where does a second target go?

| This localization model has one box slot.

Fixed tensor, variable object set

SECTION 02



How many records should the model reserve?

Zero, two, or eleven objects

No target objects

0



2



11

The annotation order does not change the target

$$\mathcal{Y} = \{(c_i, \mathbf{b}_i)\}_{i=1}^N$$

STORED ROW 1

cat + cat box

STORED ROW 2

dog + dog box

| A possible storage order.

The annotation order does not change the target

$$\mathcal{Y} = \{(c_i, \mathbf{b}_i)\}_{i=1}^N$$

STORED ROW 1

dog + dog box

STORED ROW 2

cat + cat box

| SAME TARGET SET.

A fixed tensor, a variable unordered set

NETWORK

$$f_{\theta}(x) \in \mathbb{R}^D$$

Fixed capacity

TARGET

$$\mathcal{Y} = \{y_i\}_{i=1}^N$$

Variable N

First attempt: K global slots



slot 1 slot 2 slot 3 slot 4 slot 5

Reserve five records before seeing the image.

First attempt: K global slots



slot 1	slot 2	slot 3	slot 4	slot 5
left dog	right dog	empty	empty	empty

The list order is a choice. The objects have not changed.

What if there are more objects than slots?



$$N = 11 > K = 5$$

Five records cannot represent eleven distinct instances.

Which dog is slot 1?



slot 1	slot 2	slot 3	slot 4	slot 5
left dog	right dog	empty	empty	empty

The list order is a choice. The objects have not changed.

| One valid ordering.

Which dog is slot 1?



slot 1	slot 2	slot 3	slot 4	slot 5
right dog	left dog	empty	empty	empty

The list order is a choice. The objects have not changed.

A SLOT NUMBER HAS NO INHERENT VISUAL MEANING.

Can the image give candidates an address?



Use spatial locations?

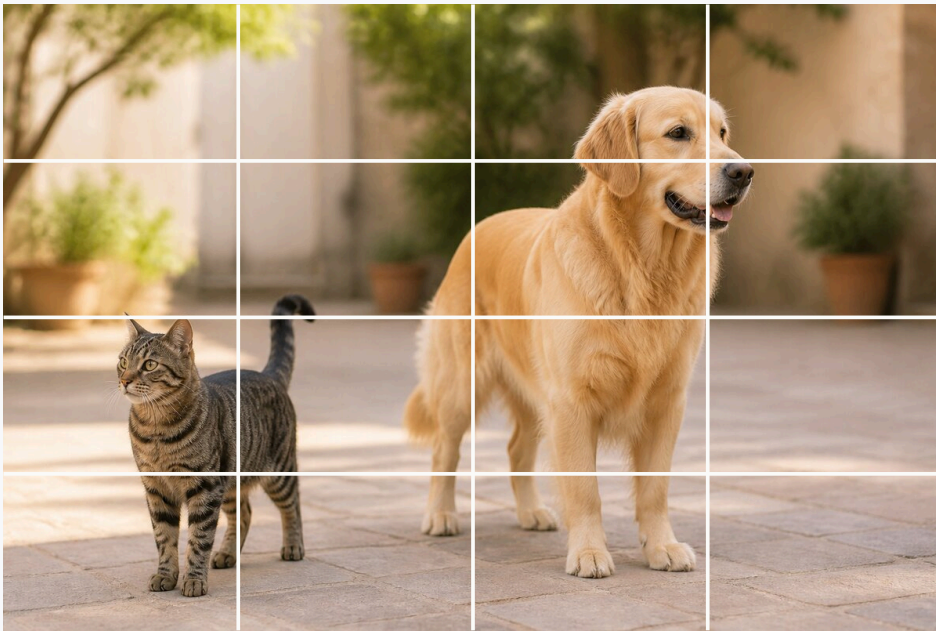
Give candidates spatial addresses

SECTION 03



Use a feature-map location to index each record.

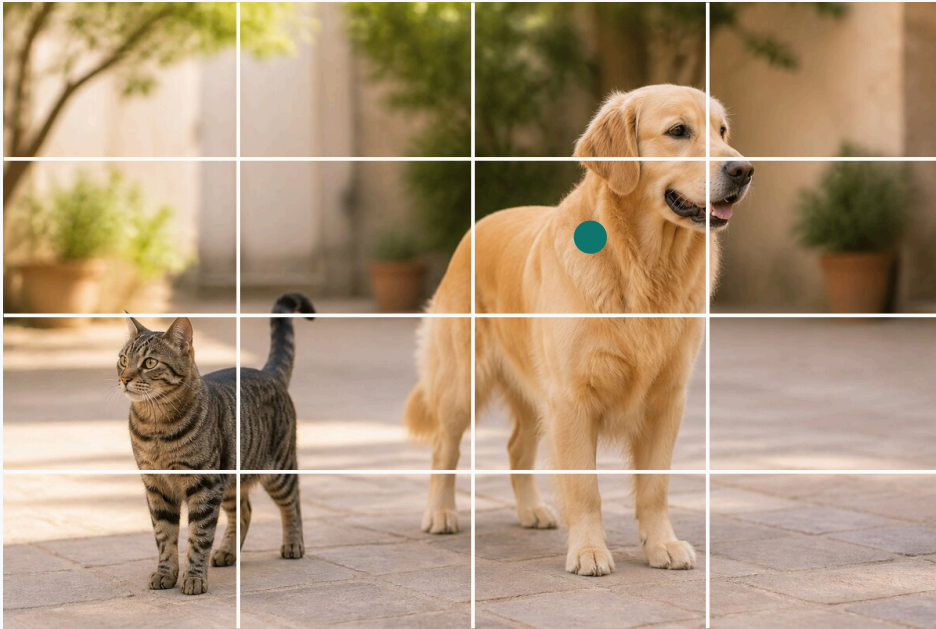
A location can index a prediction



Spatial feature grid

Keep the spatial feature grid.

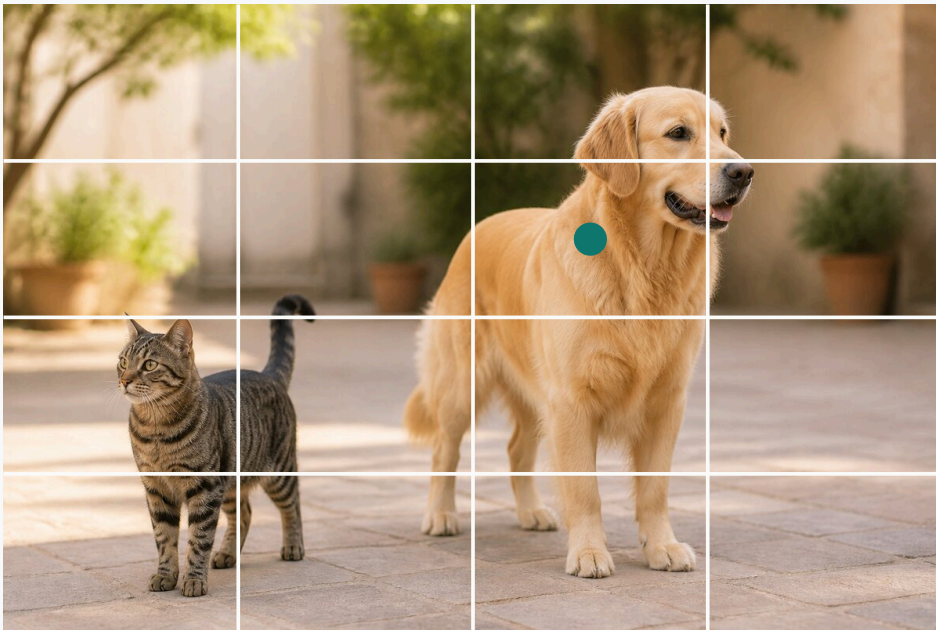
A location can index a prediction



Select one row and column.

One output address per grid location

A location can index a prediction



One output address per grid location

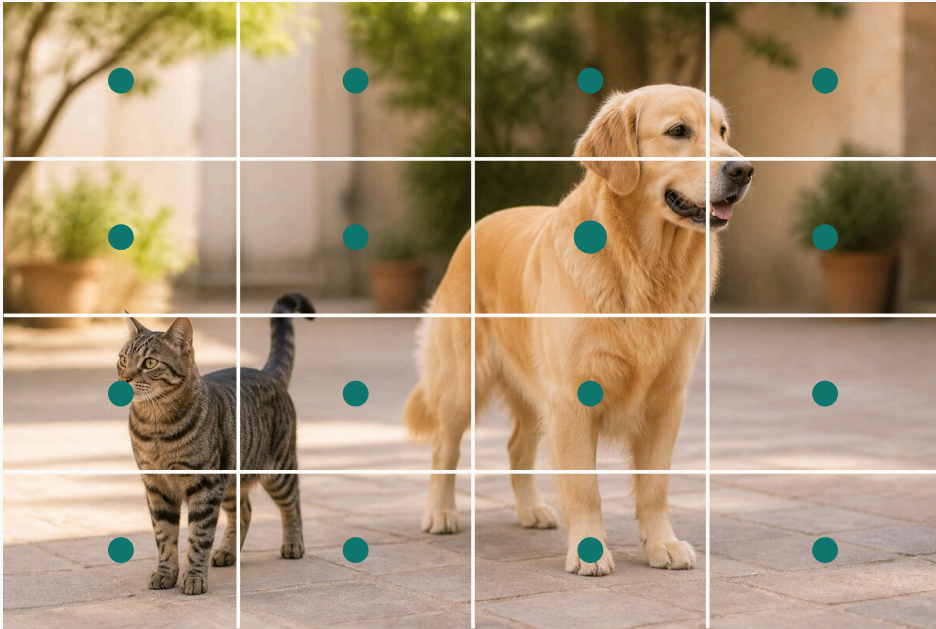
AT THIS ADDRESS

OBJECTNESS existence

BOX geometry

CLASS category

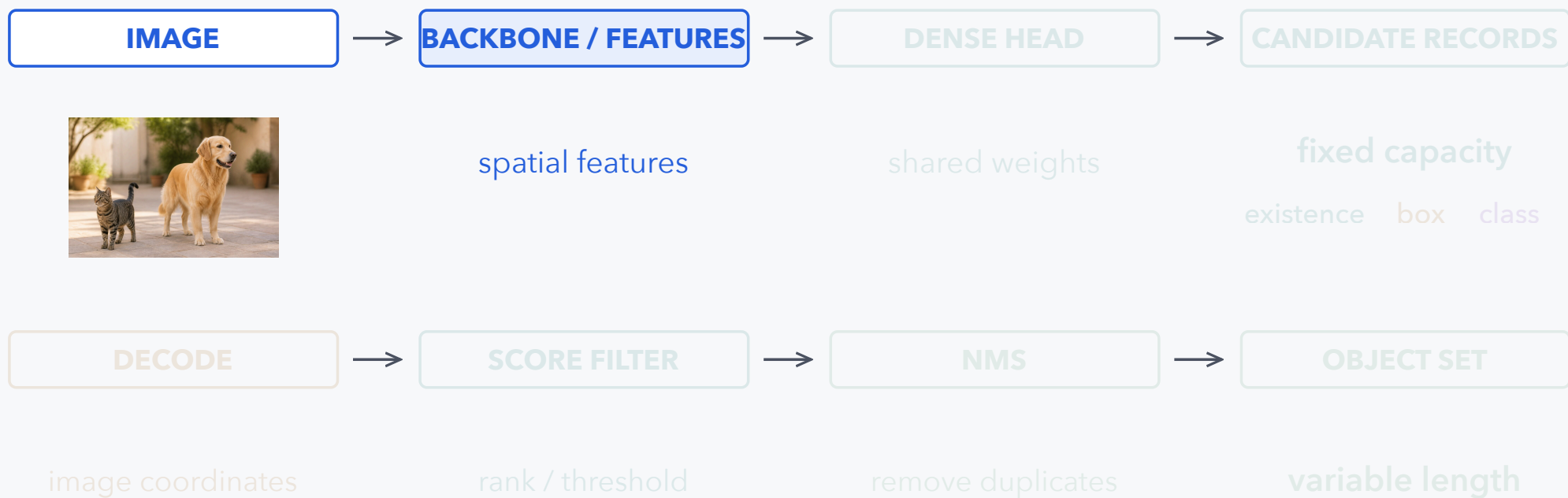
A location can index a prediction



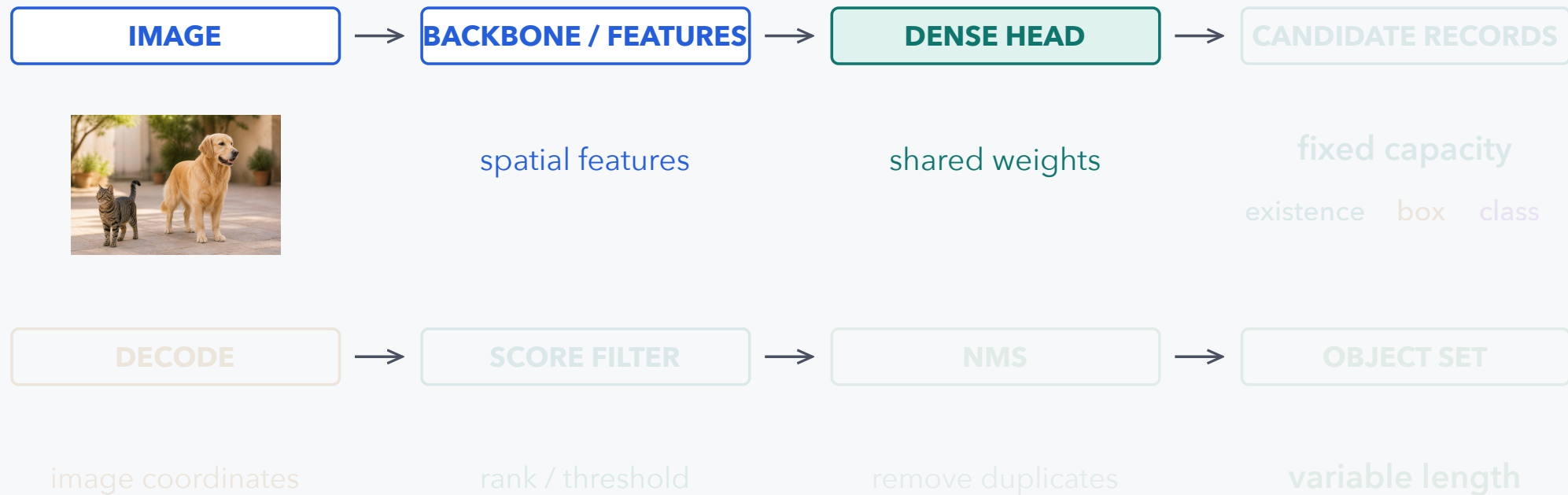
One output address per grid location

Apply the same prediction head at every location.

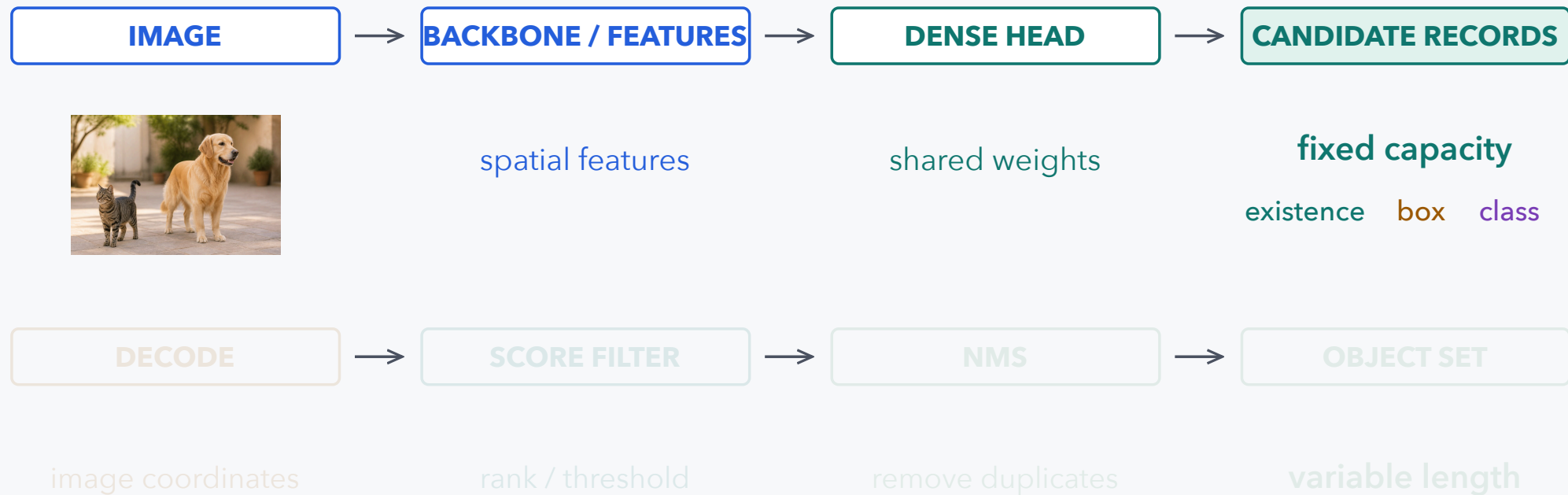
From image features to candidate records



From image features to candidate records



From image features to candidate records



One generic prediction record

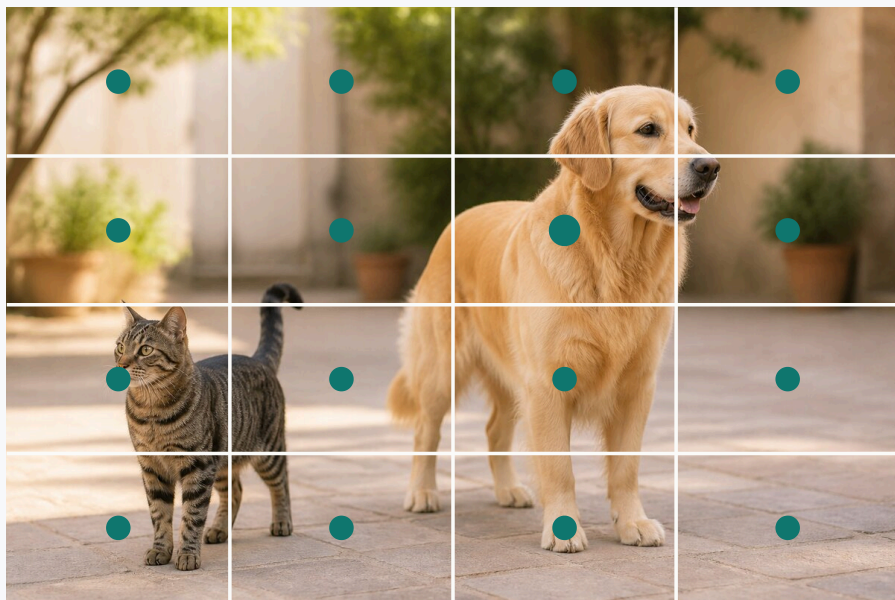
OBJECTNESS p_o

BOX (x, y, w, h)

CLASS $(p_1, \dots, p_C)$

| 1 existence value + 4 box values + C class scores.

The dimensions of the dense output



One output address per grid location

$$S \times S \times (5 + C)$$

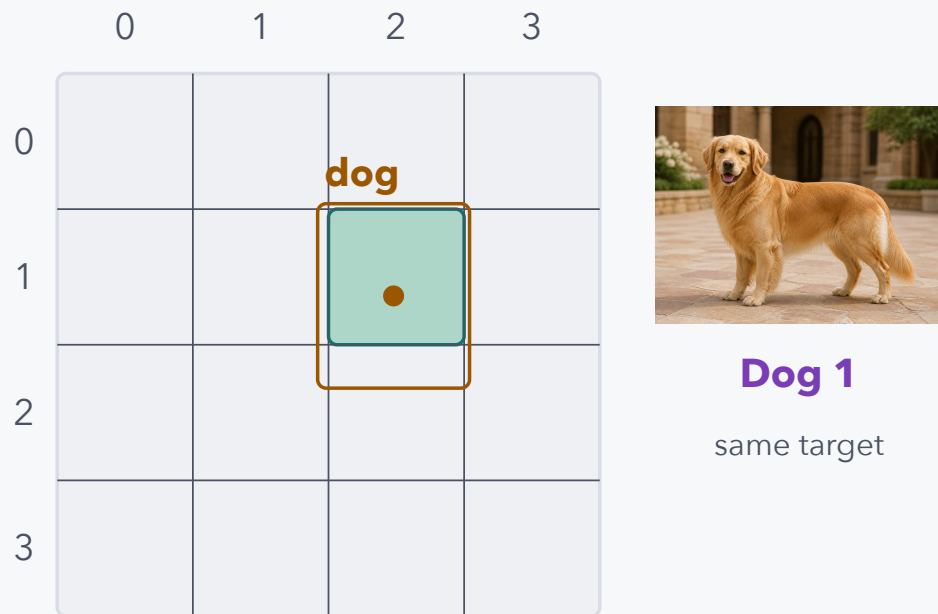
$S \times S$ addresses

C foreground classes

One record at each address

$$4 \times 4 \times 8$$

Grid cell = output address

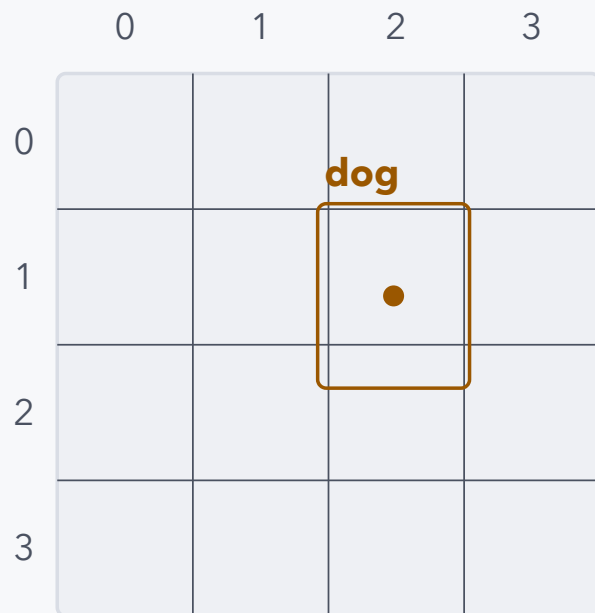


Normalized teaching grid; zero-based indices

The box can cross cell edges.

A feature at this address can see a much larger receptive field.

One dog, one normalized teaching box



Dog 1

same target

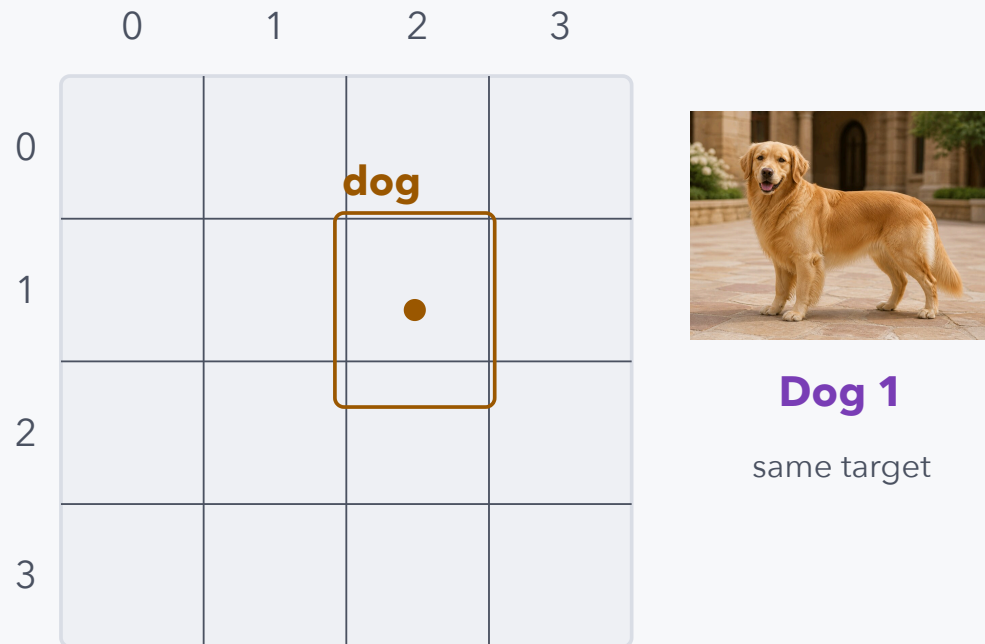
DOG 1 / GROUND TRUTH

$$(x_c, y_c, w, h) = (0.62, 0.41, 0.28, 0.34)$$

$$S = 4$$

Normalized teaching grid; zero-based indices

Which cell owns the centre?

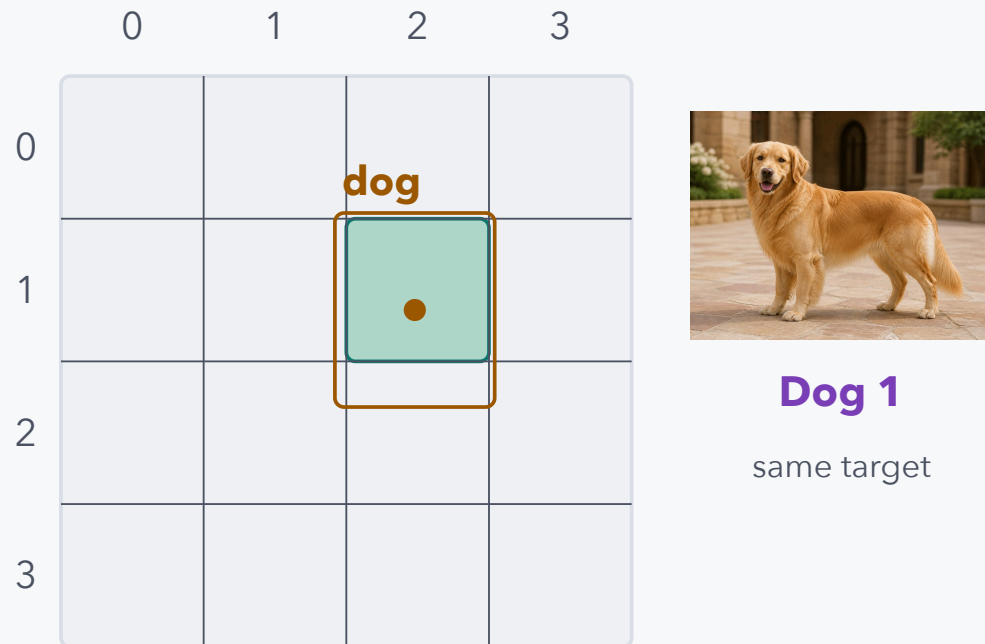


Normalized teaching grid; zero-based indices

$$j = \lfloor 4 \times 0.62 \rfloor = 2$$

Column index: 2.

Which cell owns the centre?



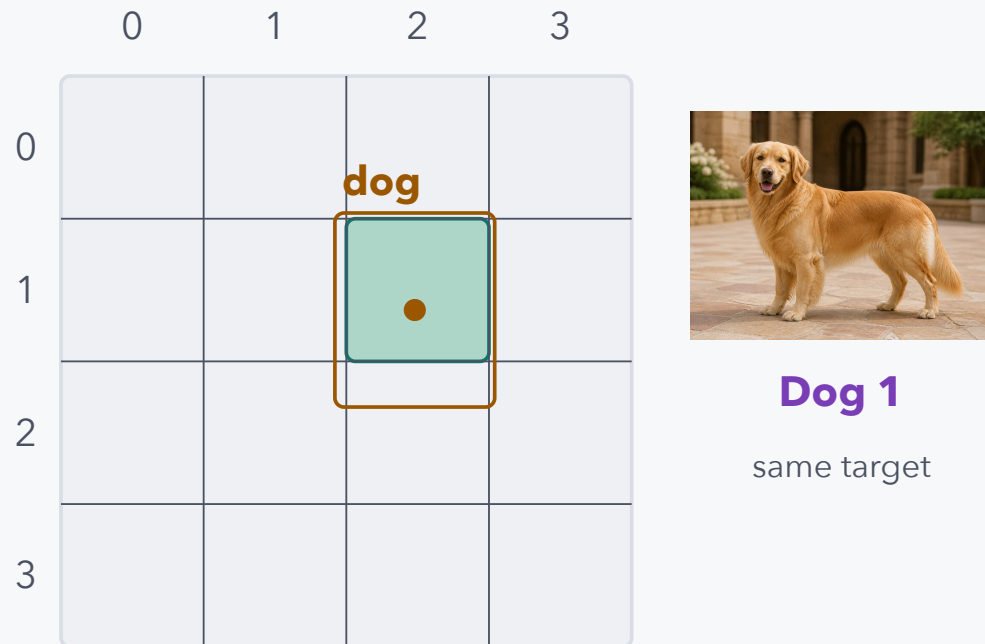
Normalized teaching grid; zero-based indices

$$j = 2$$

$$i = \lfloor 4 \times 0.41 \rfloor = 1$$

Row 1, column 2.

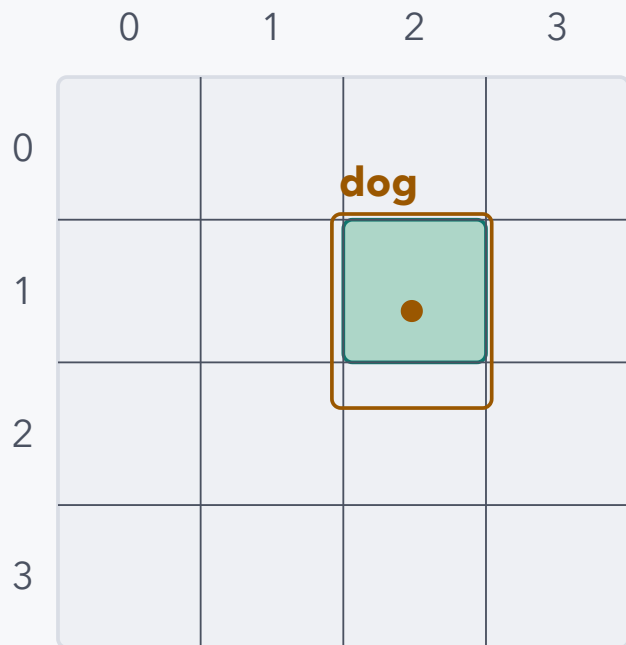
How far inside that cell is the centre?



Normalized teaching grid; zero-based indices

$$t_x = Sx_c - j$$
$$= 4 \times 0.62 - 2 = 0.48$$

How far inside that cell is the centre?



Dog 1

same target

$$t_x = 0.48$$

$$t_y = Sy_c - i$$

$$= 4 \times 0.41 - 1 = 0.64$$

Normalized teaching grid; zero-based indices

Construct the target record

TARGET AT ROW 1, COLUMN 2

OBJECTNESS

1

BOX

(0.48, 0.64, 0.28, 0.34)

CLASS

dog

| Local centre offsets; full-image normalized width and height.

The model emits a record at the same address

ILLUSTRATIVE TEACHING PREDICTION

OBJECTNESS

0.80

BOX

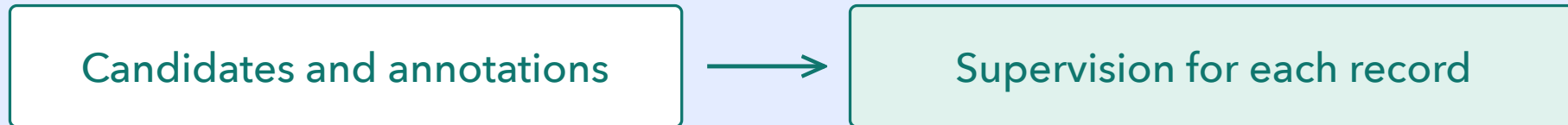
(0.45, 0.60, 0.30, 0.32)

CLASS

cat 0.10
dog 0.85
other 0.05

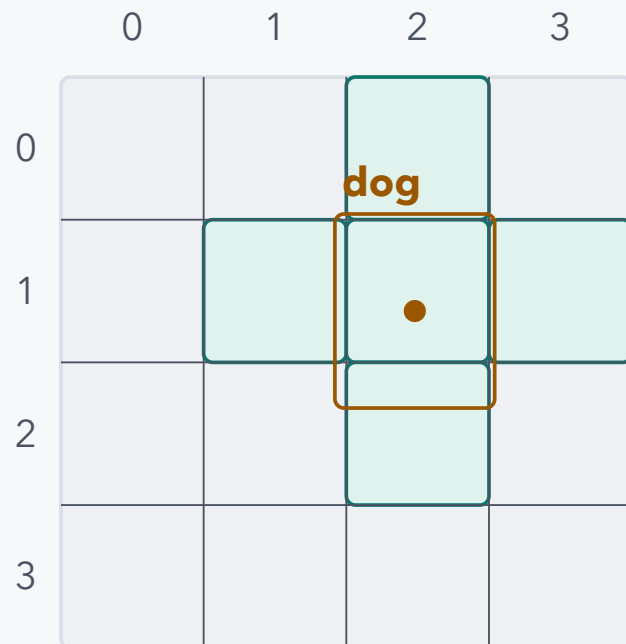
Which candidate should learn which object?

SECTION 04



A loss needs a target for each prediction.

Which of these candidates should learn the dog?



Dog 1

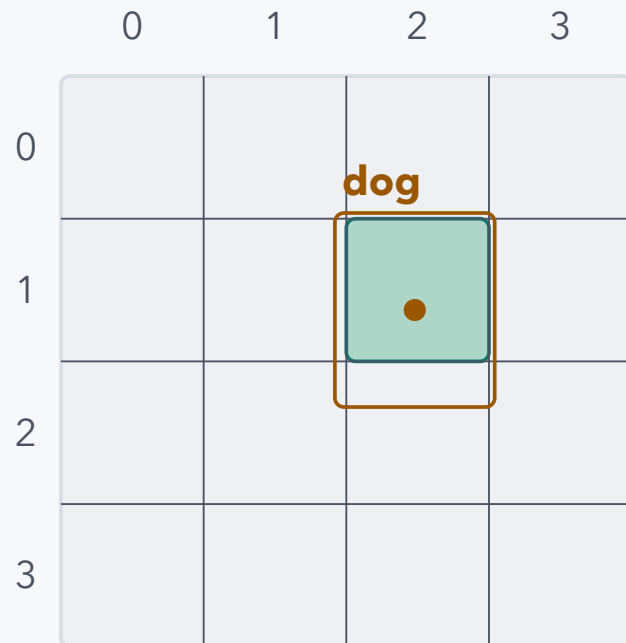
same target

Several nearby candidates see it.

Which one receives its target?

Normalized teaching grid; zero-based indices

Which of these candidates should learn the dog?



Dog 1

same target

TEACHING ASSIGNMENT RULE

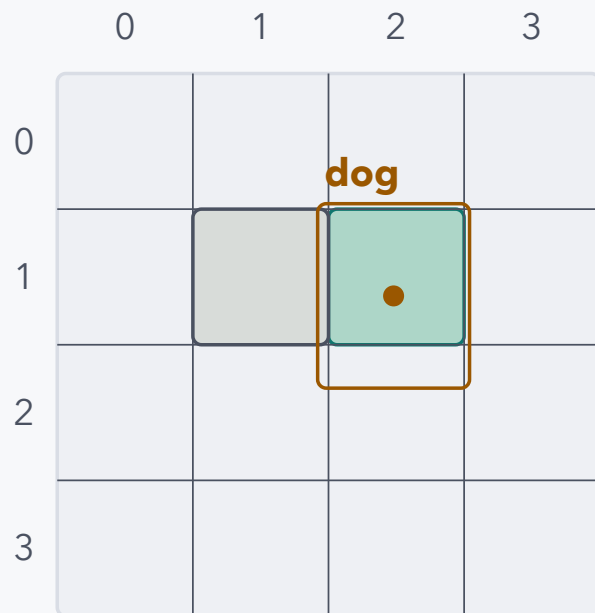
Assign the object to the cell containing its centre.

$$(i, j) = (1, 2)$$

Modern detectors use model-specific assignment rules.

Normalized teaching grid; zero-based indices

Positive, background, and ignored candidates



Dog 1

same target

POSITIVE

The assigned cell learns the dog record.

BACKGROUND

An unassigned cell learns objectness 0.

IGNORE, WHEN USED

An ignored candidate contributes no loss.

Normalized teaching grid; zero-based indices

Objectness: is an object assigned here?

TARGET

$$y_o = 1$$

PREDICTION

$$p_o = 0.80$$

Objectness: is an object assigned here?

TARGET / PREDICTION

1 / 0.80

$$\begin{aligned}\mathcal{L}_{obj} &= -\log p_o \\ &= -\log(0.80) \approx 0.223\end{aligned}$$

Class: how much probability went to dog?

TARGET CLASS

dog

PREDICTED CLASS SCORES

cat: 0.10

dog: 0.85

other: 0.05

Class: how much probability went to dog?

CORRECT CLASS

$$p_{dog} = 0.85$$

$$\begin{aligned}\mathcal{L}_{cls} &= -\log p_{dog} \\ &\approx 0.163\end{aligned}$$

Box: compare one coordinate at a time

SIMPLE TEACHING L1 LOSS / SUM REDUCTION

Target

(0.48, 0.64, 0.28, 0.34)

Prediction

(0.45, 0.60, 0.30, 0.32)

$$|\hat{t}_x - t_x| = |0.45 - 0.48| = 0.03$$

| 1 of 4 coordinate differences

Box: compare one coordinate at a time

SIMPLE TEACHING L1 LOSS / SUM REDUCTION

Target

(0.48, 0.64, 0.28, 0.34)

Prediction

(0.45, 0.60, 0.30, 0.32)

$$|\hat{t}_y - t_y| = |0.60 - 0.64| = 0.04$$

| 2 of 4 coordinate differences

Box: compare one coordinate at a time

SIMPLE TEACHING L1 LOSS / SUM REDUCTION

Target

(0.48, 0.64, 0.28, 0.34)

Prediction

(0.45, 0.60, 0.30, 0.32)

$$|\hat{w} - w| = |0.30 - 0.28| = 0.02$$

| 3 of 4 coordinate differences

Box: compare one coordinate at a time

SIMPLE TEACHING L1 LOSS / SUM REDUCTION

Target

(0.48, 0.64, 0.28, 0.34)

Prediction

(0.45, 0.60, 0.30, 0.32)

$$|\hat{h} - h| = |0.32 - 0.34| = 0.02$$

| 4 of 4 coordinate differences

Box: compare one coordinate at a time

SIMPLE TEACHING L1 LOSS / SUM REDUCTION

Target

(0.48, 0.64, 0.28, 0.34)

Prediction

(0.45, 0.60, 0.30, 0.32)

$$\mathcal{L}_{box} = \sum_{k=1}^4 |\hat{b}_k - b_k|$$

$$= 0.03 + 0.04 + 0.02 + 0.02 = 0.110$$

| Real detectors often use IoU-family losses or other box parameterizations.

Add the three supervised terms

$$\mathcal{L} = \lambda_{obj} \mathcal{L}_{obj} + \lambda_{box} \mathcal{L}_{box} + \lambda_{cls} \mathcal{L}_{cls}$$

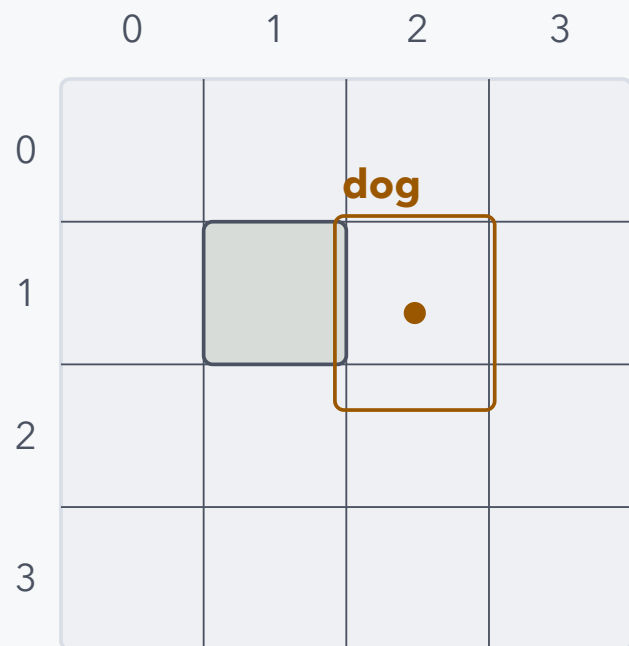
For this worksheet, set all three weights to 1.

Add the three supervised terms

$$\mathcal{L} = 0.223 + 0.110 + 0.163 \approx 0.496$$

Assignment determines which targets enter these losses.

A nearby unassigned candidate



Dog 1

same target

ROW 1, COLUMN 1

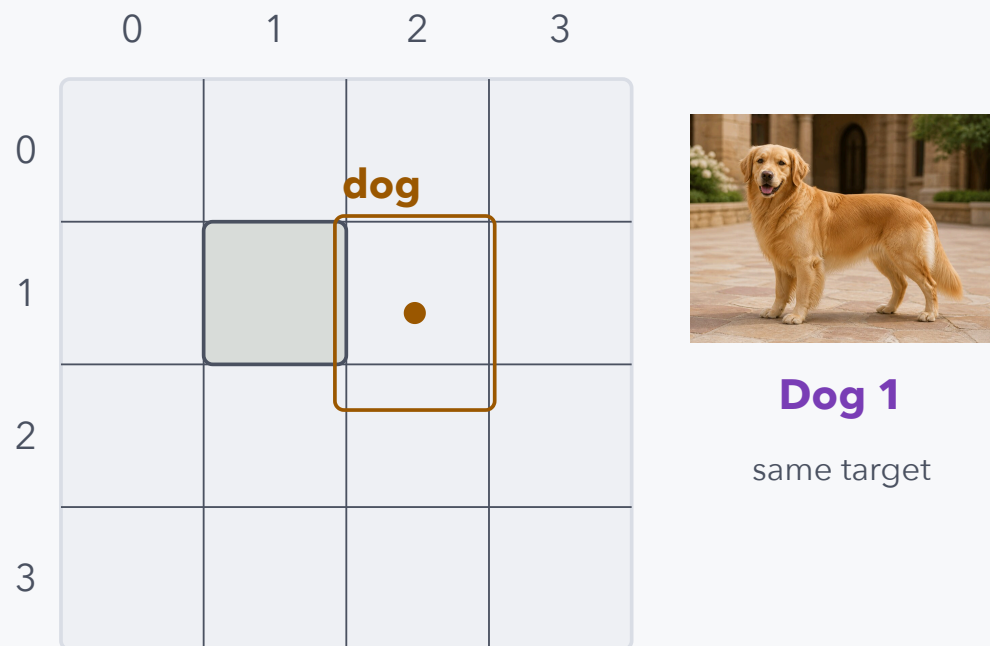
$$y_o = 0$$

$$p_o = 0.20$$

No dog class or box target.

Normalized teaching grid; zero-based indices

A nearby unassigned candidate



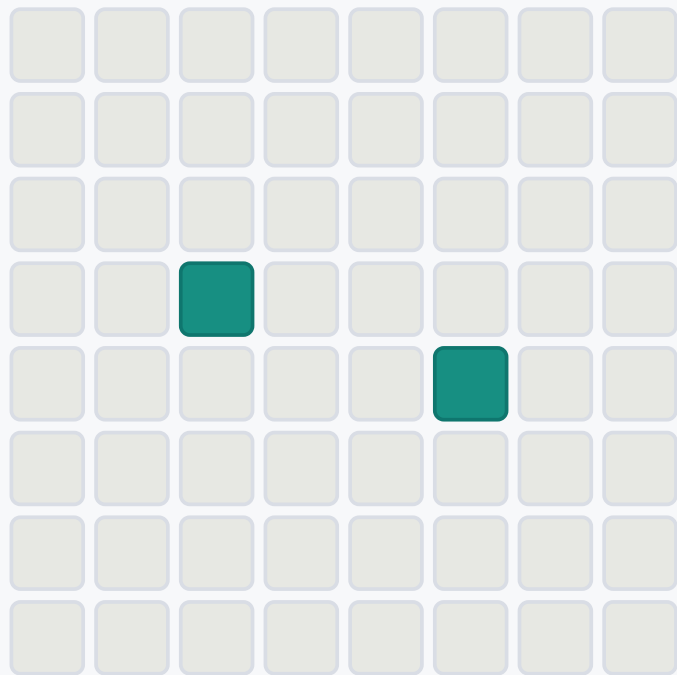
Normalized teaching grid; zero-based indices

$$\mathcal{L}_{obj} = -\log(1 - p_o)$$

$$= -\log(0.80) \approx 0.223$$

The box and class terms are masked out.

Many background terms can dominate a sum



Separate 8 × 8 example

64 candidates
2 positive
62 background

Many background terms can dominate a sum

$$\sum_{i=1}^2 \mathcal{L}_i^+$$

2 positive terms

$$\sum_{j=1}^{62} \mathcal{L}_j^-$$

62 background terms

| Weighting, focal loss, or hard-example selection can reduce this imbalance.

A teaching record, not a universal detector format

OBJECTNESS

objectness

BOX

box

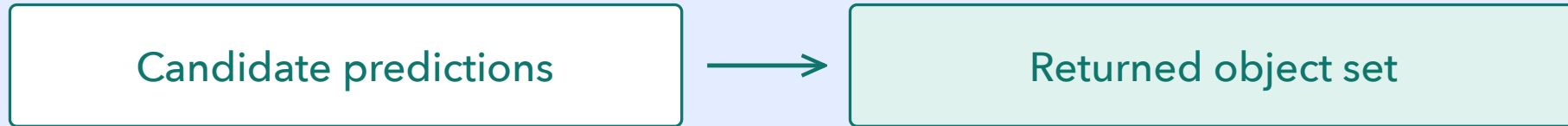
CLASS

class

The three fields make the derivation concrete. Modern detectors can factor confidence, classes, and geometry differently.

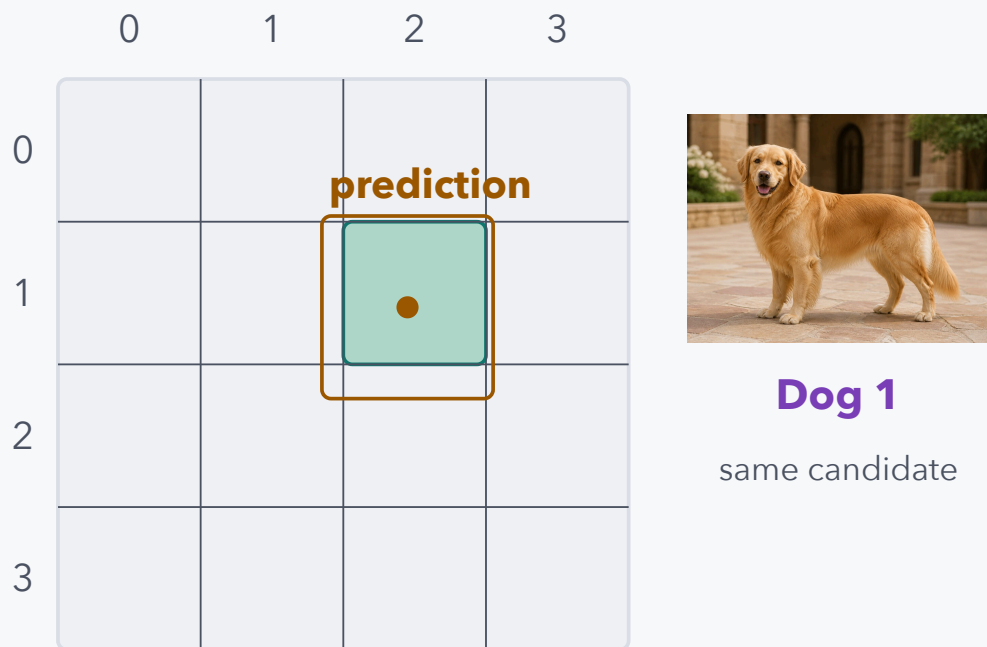
From candidates to final detections

SECTION 05



At inference there are no annotations to consult.

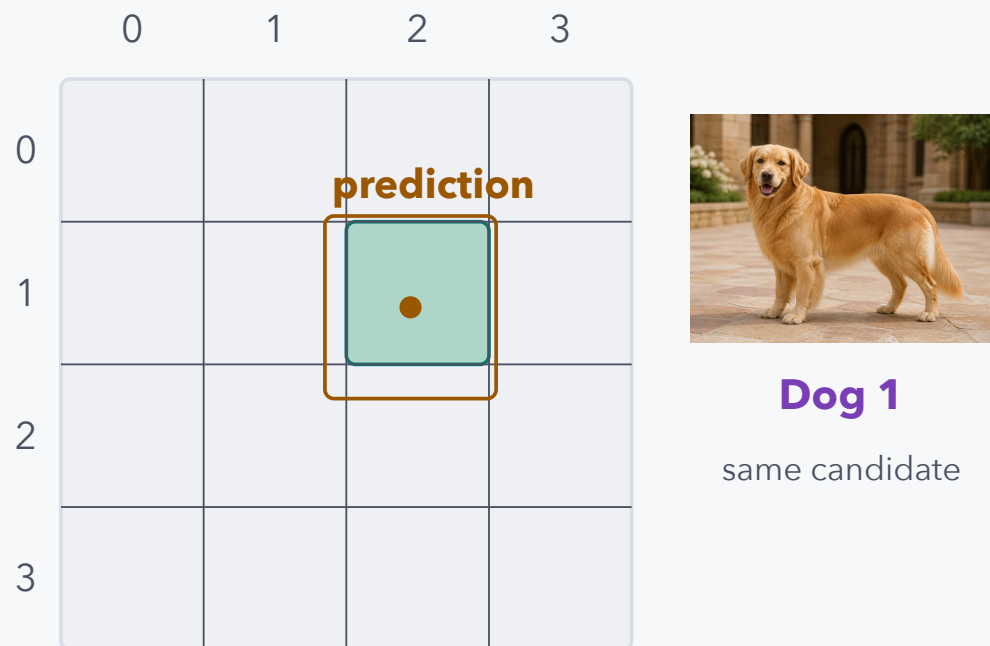
Decode the same predicted box



Normalized teaching grid; zero-based indices

$$\hat{x}_c = \frac{j + \hat{t}_x}{S} = \frac{2 + 0.45}{4} = 0.6125$$

Decode the same predicted box



Normalized teaching grid; zero-based indices

$$\hat{y}_c = \frac{i + \hat{t}_y}{S} = \frac{1 + 0.60}{4} = 0.4000$$

Decode the same predicted box

DECODED PREDICTION

$$(\hat{x}_c, \hat{y}_c, \hat{w}, \hat{h}) = \\ (0.6125, 0.4000, 0.3000, 0.3200)$$

| Width and height were already normalized to the full image.

A score for this predicted dog

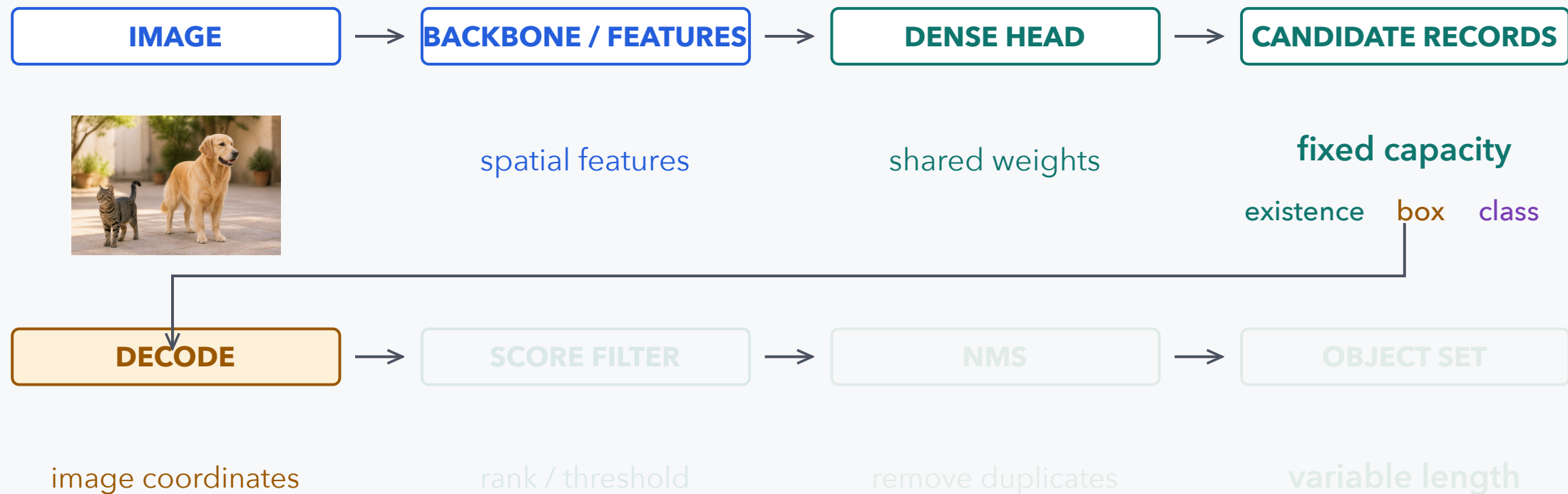
ONE DECLARED SCORING CONVENTION

$$s_{dog} = p_o p(dog \mid object)$$

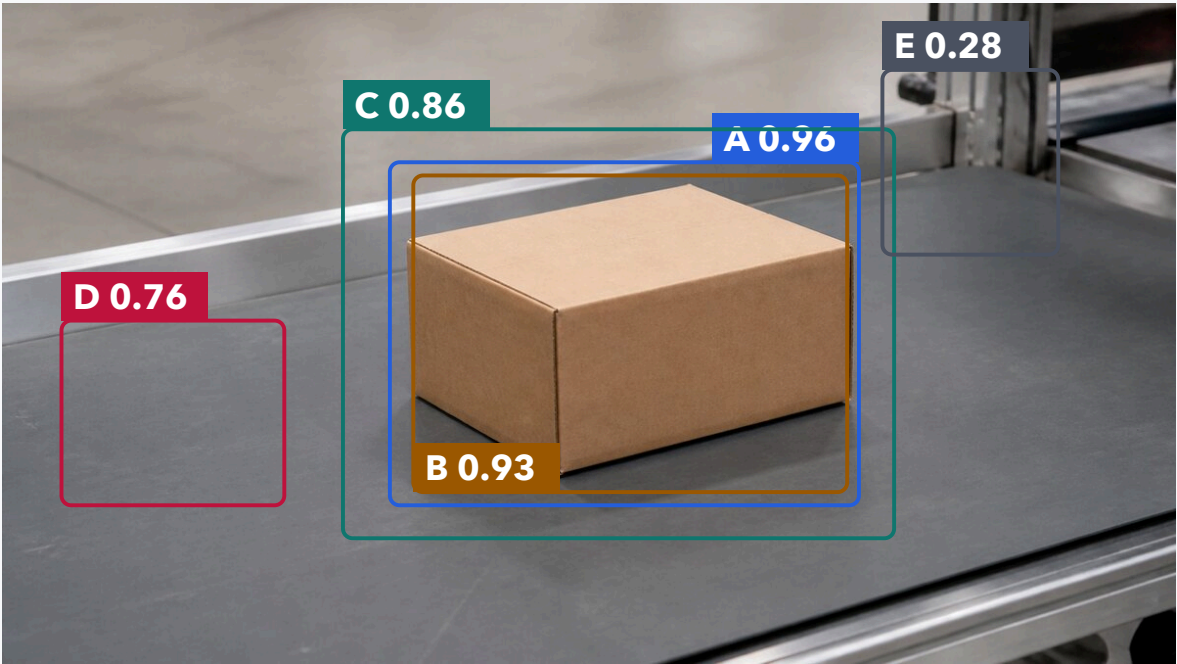
$$0.80 \times 0.85 = 0.68$$

The score ranks this candidate.

Decoding interprets the fixed candidate tensor

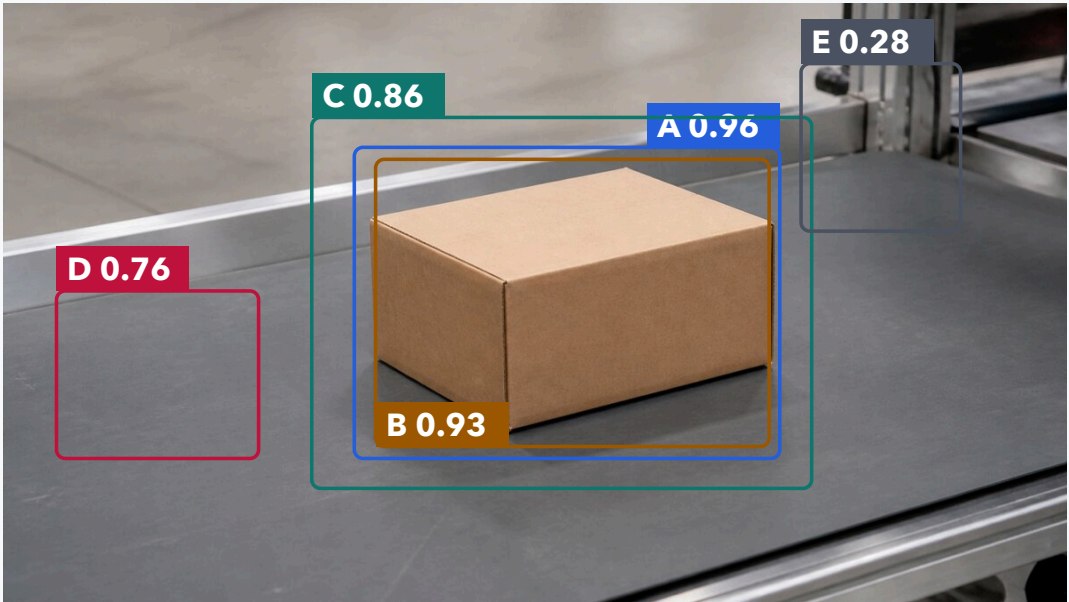


Now several candidates describe one parcel



CANDIDATE	SCORE	STATUS
A	0.96	candidate
B	0.93	candidate
C	0.86	candidate
D	0.76	candidate
E	0.28	candidate

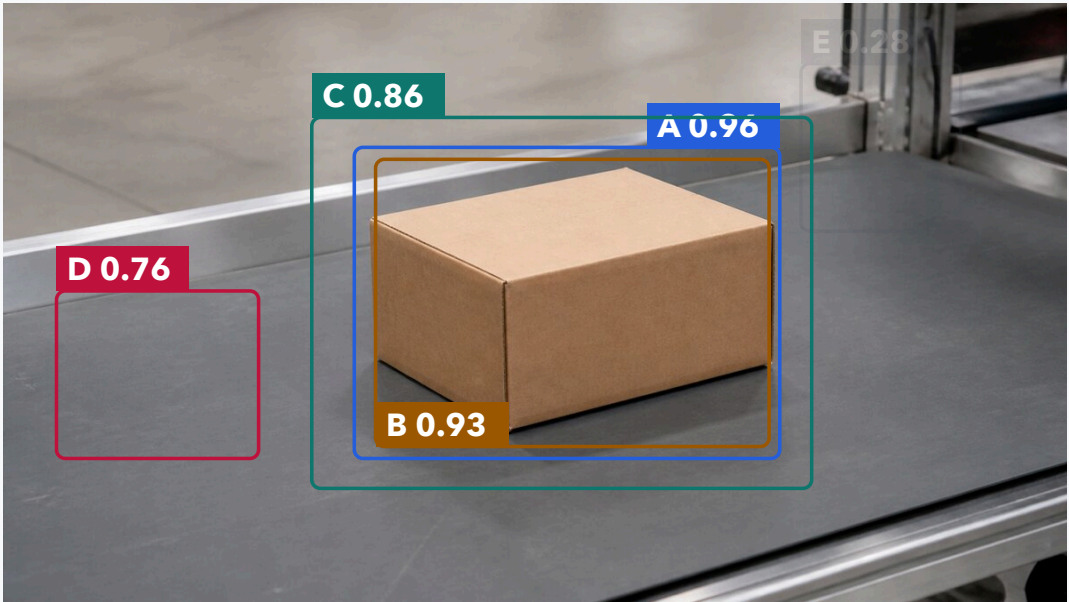
Filter by score



CANDIDATE	SCORE	STATUS
A	0.96	candidate
B	0.93	candidate
C	0.86	candidate
D	0.76	candidate
E	0.28	candidate

Retain scores at least 0.70.

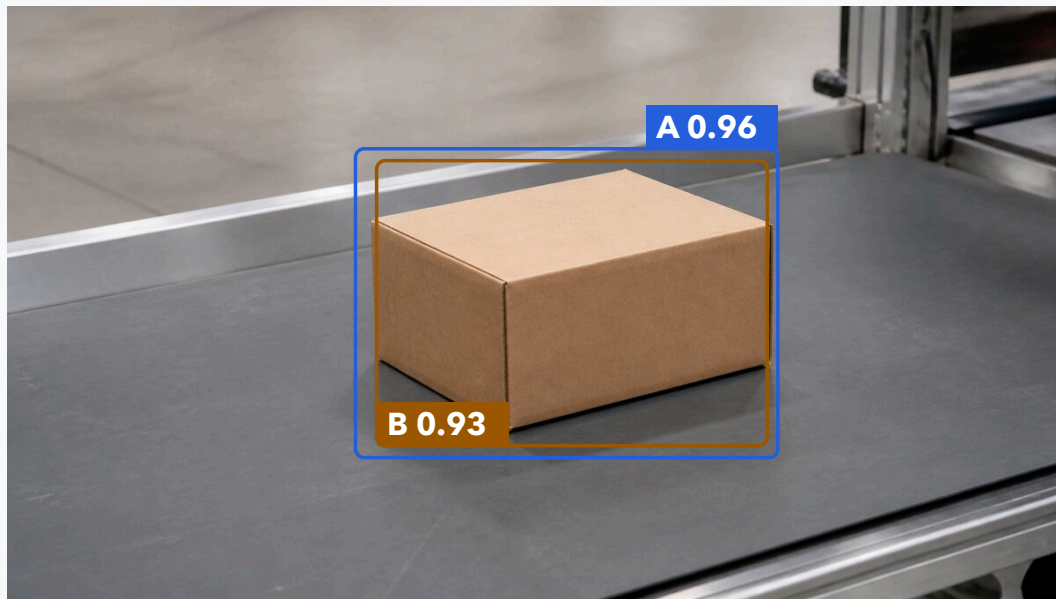
Filter by score



CANDIDATE	SCORE	STATUS
A	0.96	candidate
B	0.93	candidate
C	0.86	candidate
D	0.76	candidate
E	0.28	filtered

E fades. Why are four boxes still left for one parcel?

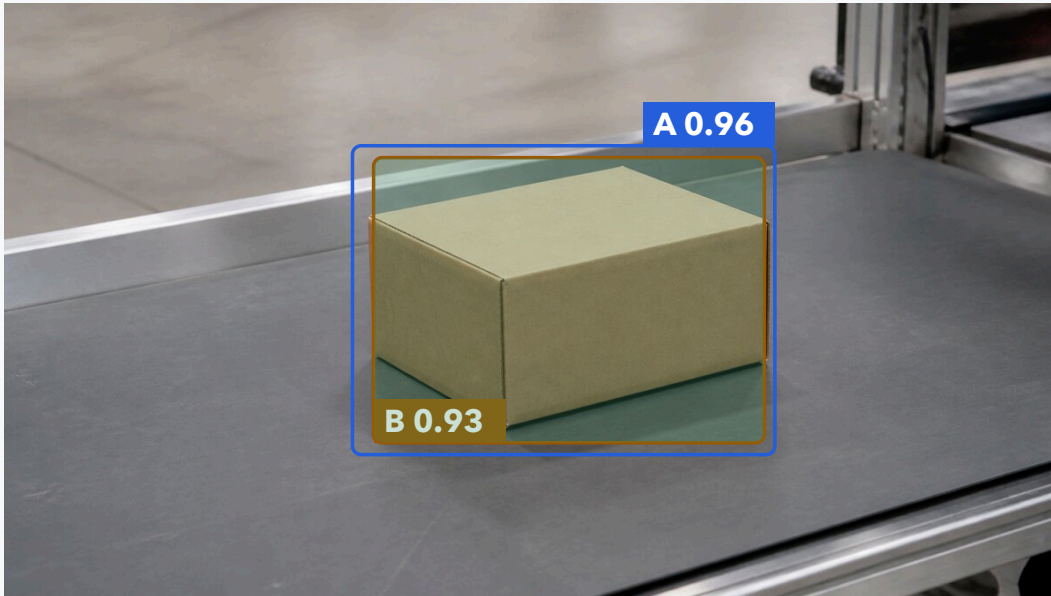
How much do A and B overlap?



$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Shared area divided by total covered area.

How much do A and B overlap?



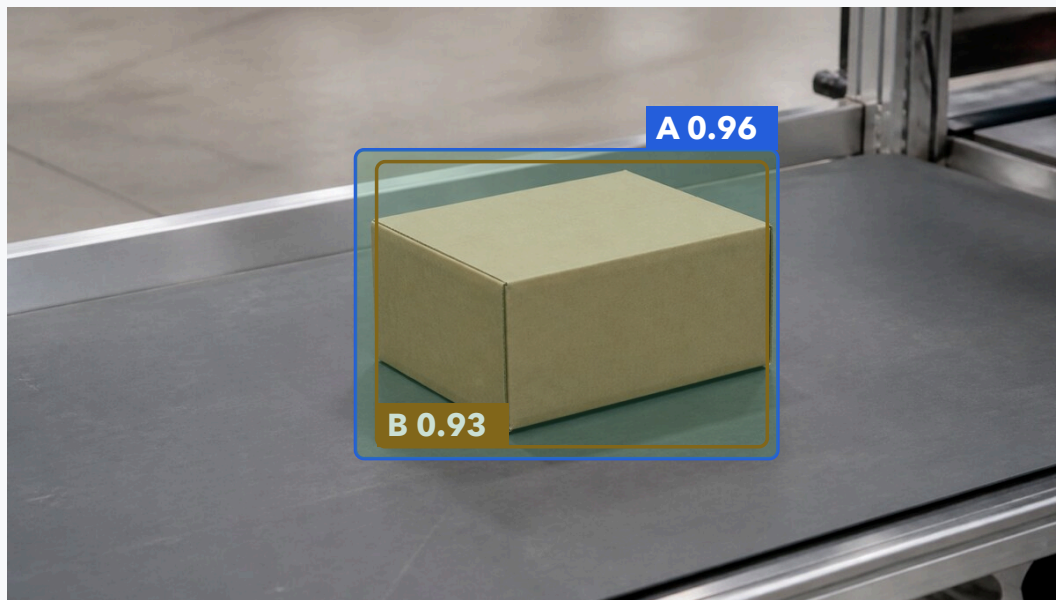
Shaded intersection: A and B

$$|A| = 0.2080$$

$$|B| = 0.1776$$

$$|A \cap B| = 0.1776$$

How much do A and B overlap?



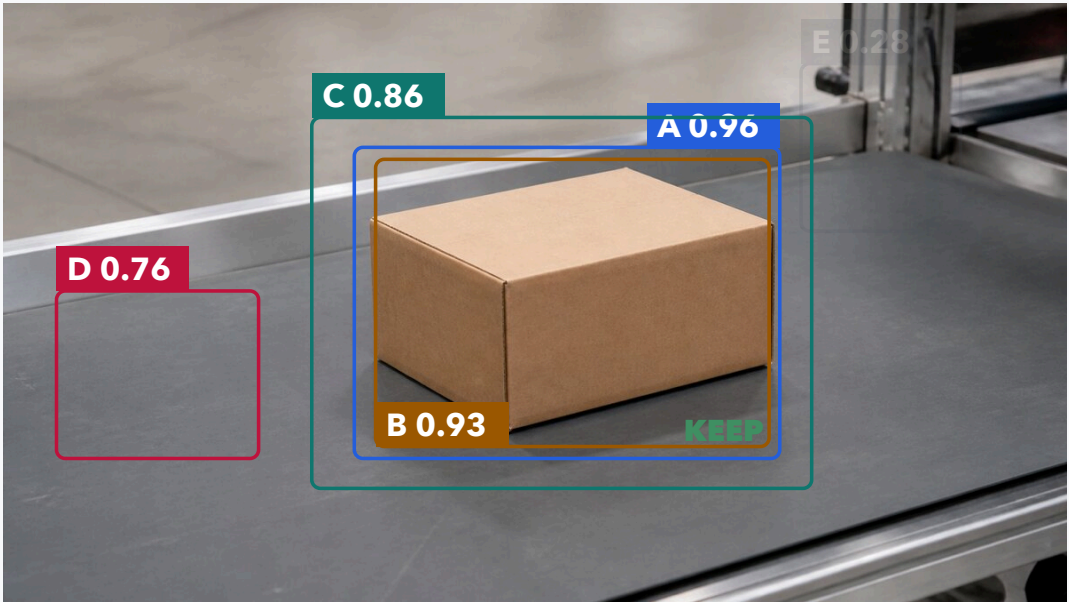
Shaded union: B lies inside A

Union = area A + area B – intersection

$$|A \cup B| = 0.2080$$

$$\text{IoU}(A, B) = \frac{0.1776}{0.2080} \approx 0.854$$

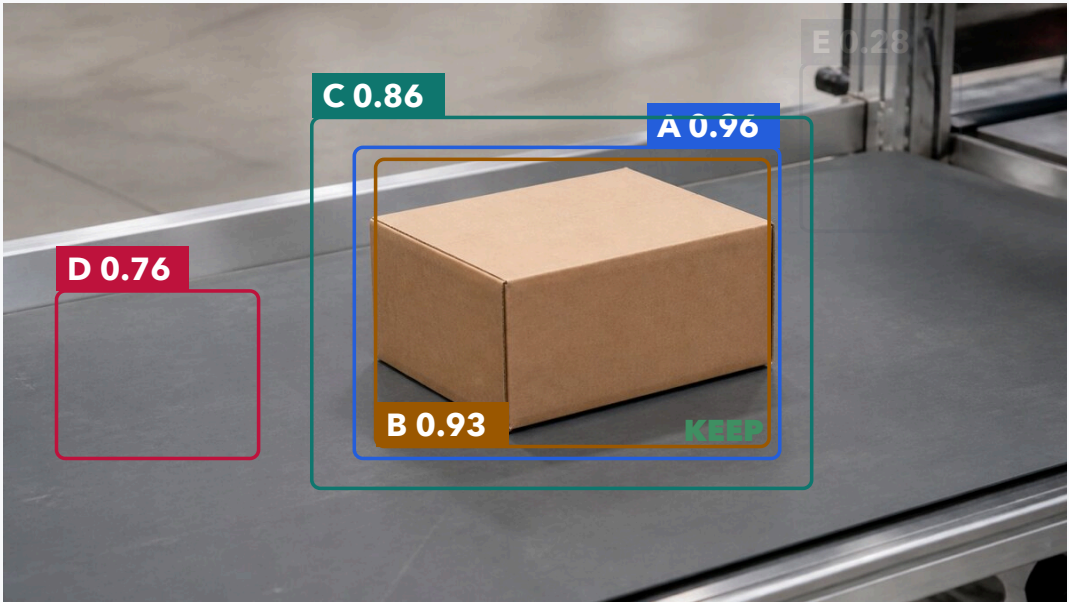
Non-maximum suppression



CANDIDATE	SCORE	STATUS
A	0.96	KEEP
B	0.93	pending
C	0.86	pending
D	0.76	pending
E	0.28	filtered

- 1. Keep A, the highest-scoring pending candidate.

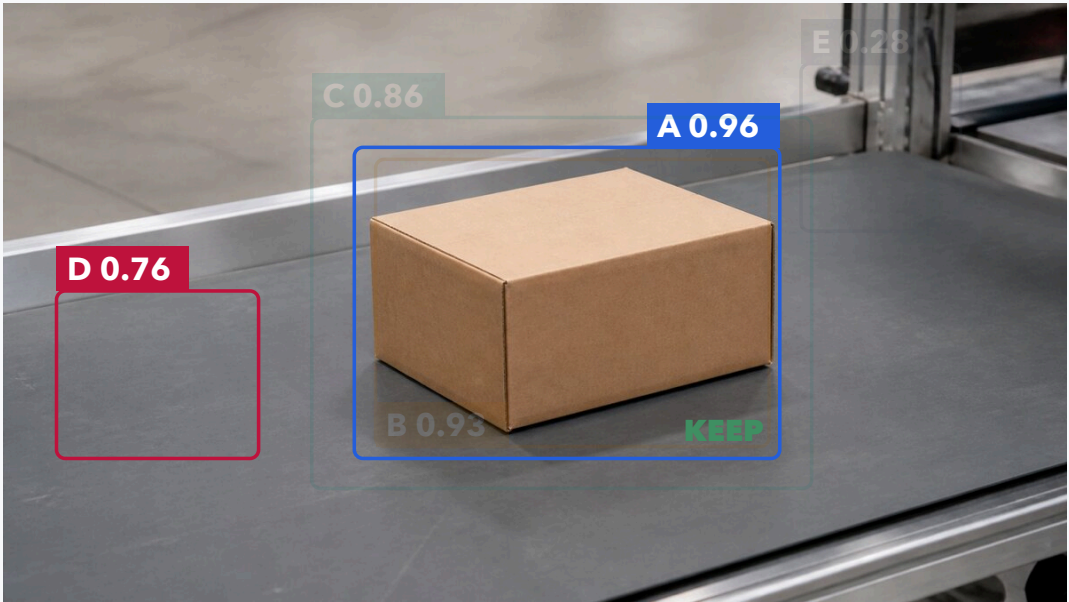
Non-maximum suppression



CANDIDATE	SCORE	STATUS
A	0.96	KEEP
B	0.93	IoU 0.85
C	0.86	IoU 0.71
D	0.76	IoU 0.00
E	0.28	filtered

2. Compare A with the other pending boxes.

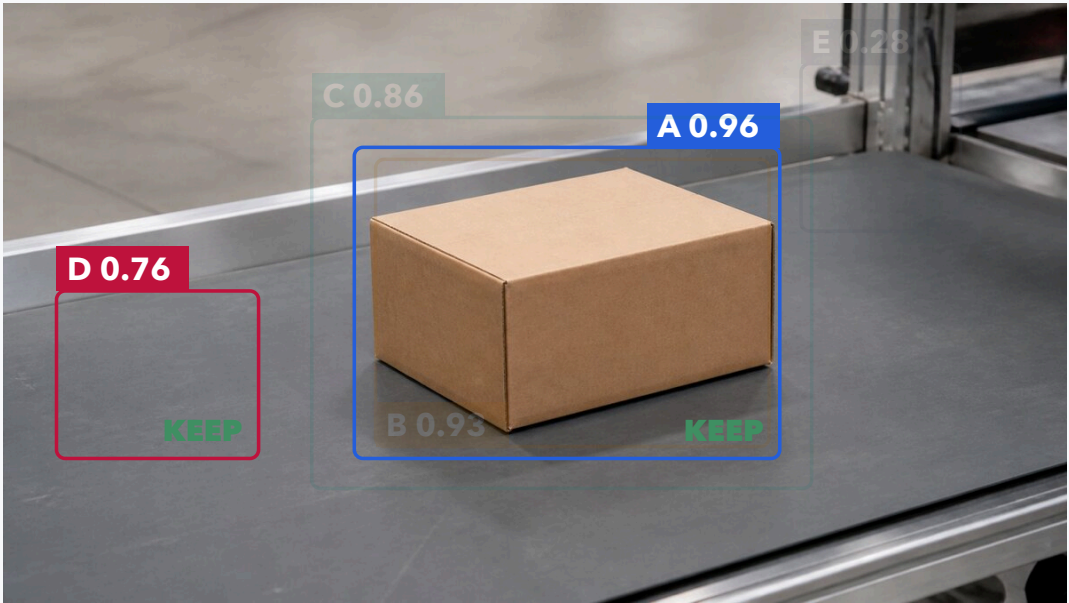
Non-maximum suppression



CANDIDATE	SCORE	STATUS
A	0.96	KEEP
B	0.93	suppress
C	0.86	suppress
D	0.76	pending
E	0.28	filtered

3. Suppress B and C: IoU exceeds 0.50. D stays pending.

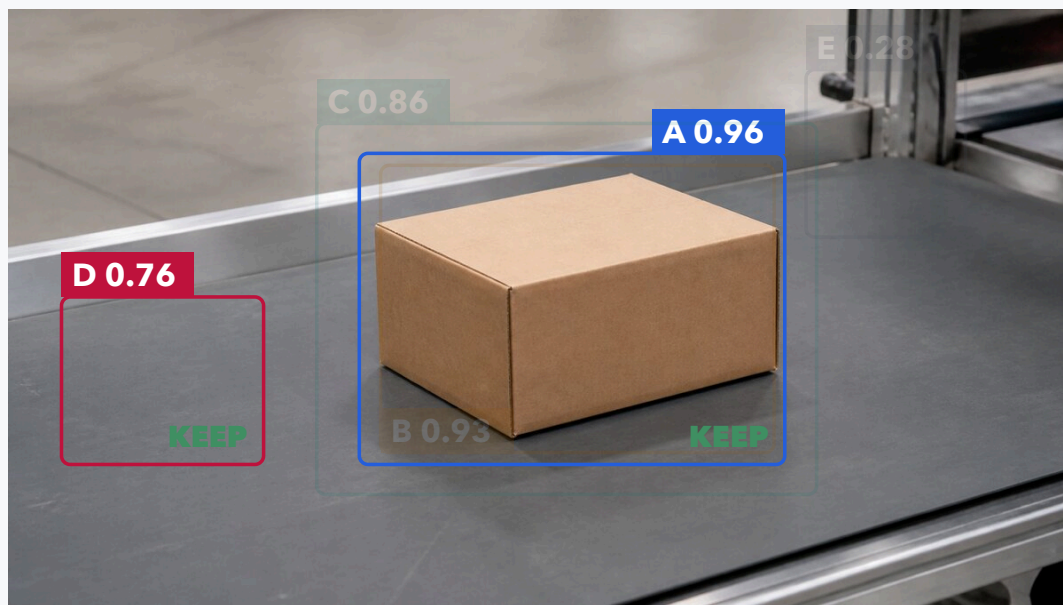
Non-maximum suppression



CANDIDATE	SCORE	STATUS
A	0.96	KEEP
B	0.93	suppress
C	0.86	suppress
D	0.76	KEEP
E	0.28	filtered

4. Keep D. The pending set is empty.

The returned set has variable length

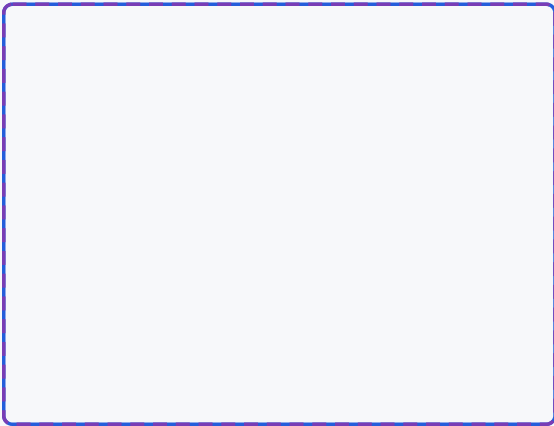


CANDIDATE	SCORE	STATUS
A	0.96	KEEP
B	0.93	suppress
C	0.86	suppress
D	0.76	KEEP
E	0.28	filtered

5 candidates → 4 after filtering → 2 after NMS.

Class-aware NMS separates class groups

dog



cat

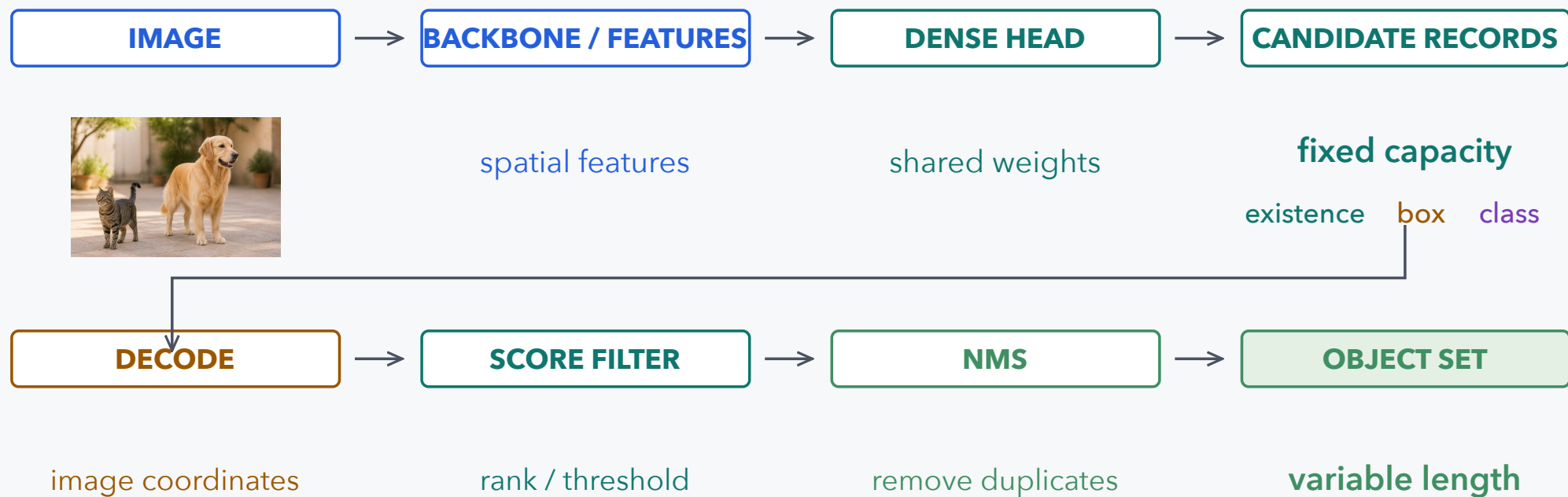
Same geometry.

Different predicted classes.

Class-aware NMS does not make these boxes compete.

Some systems use class-agnostic NMS.

Fixed capacity, scene-dependent returned set



FIXED CANDIDATE CAPACITY → VARIABLE RETURNED SET.

Three different matching problems

TRAINING

candidate $\leftrightarrow$ ground truth

assign supervision

INFERENCE

prediction $\leftrightarrow$ prediction

remove duplicates

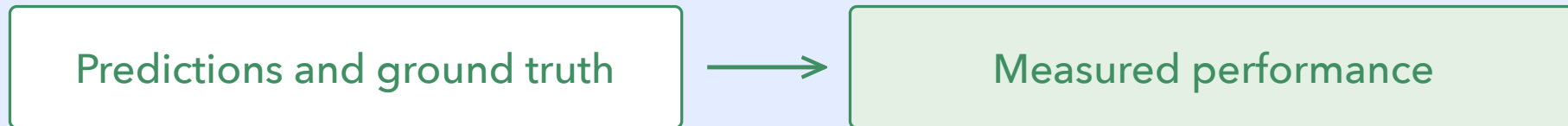
EVALUATION

prediction $\leftrightarrow$ ground truth

measure correctness

How do we measure a detector?

SECTION 06



Compare the returned predictions with annotated instances.

When does a prediction count as a true positive?

- Same image
- Same class
- IoU at or above the evaluation threshold
- Ground-truth instance not already claimed

Precision and recall use different denominators

$$P = \frac{TP}{TP + FP}$$

Among the predictions, how many are correct?

$$P = \frac{2}{2 + 1} = \frac{2}{3}$$

Precision and recall use different denominators

$$R = \frac{TP}{TP + FN}$$

Among the annotated objects, how many did we find?

$$R = \frac{2}{2 + 0} = 1$$

A deployment cutoff and AP serve different purposes

DEPLOYMENT SCORE THRESHOLD

Choose which predictions the system returns at one operating point.

AP EVALUATION

Use the ranked prediction list across operating points.

A ranked prefix traces the PR curve

TWO GROUND-TRUTH OBJECTS

P1	TP	0.95
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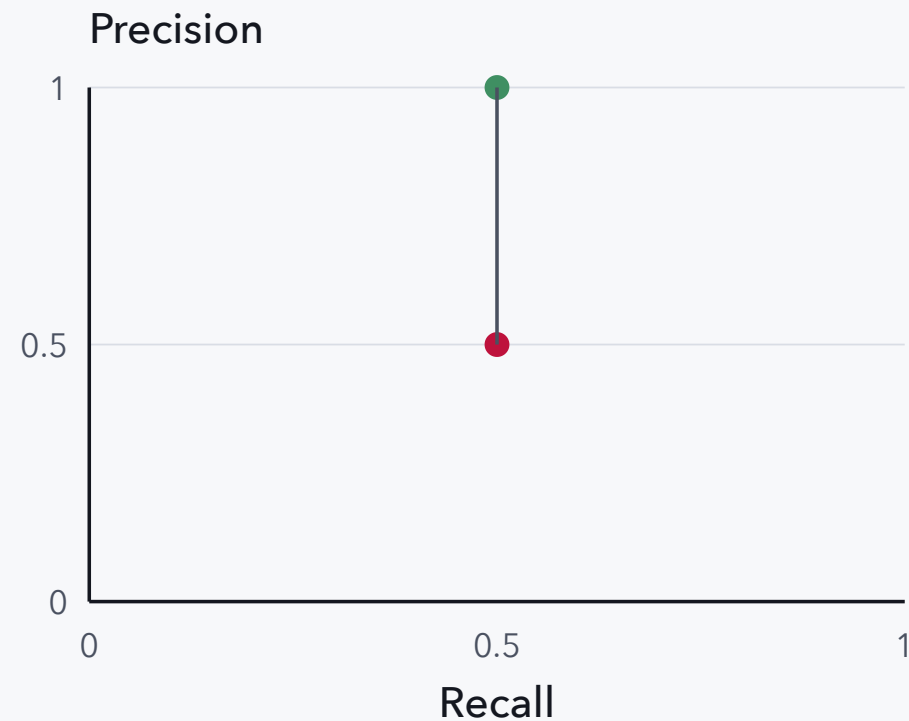


After 1: precision 1.000, recall 0.500.

A ranked prefix traces the PR curve

TWO GROUND-TRUTH OBJECTS

P1	TP	0.95
P2	FP	0.89

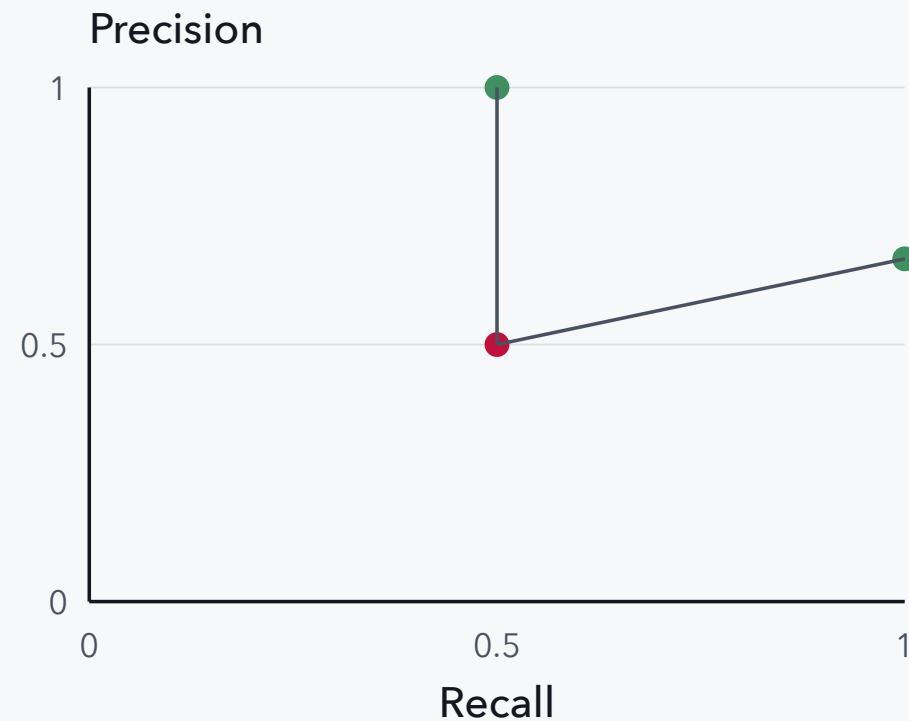


After 2: precision 0.500, recall 0.500.

A ranked prefix traces the PR curve

TWO GROUND-TRUTH OBJECTS

P1	TP	0.95
P2	FP	0.89
P3	TP	0.88

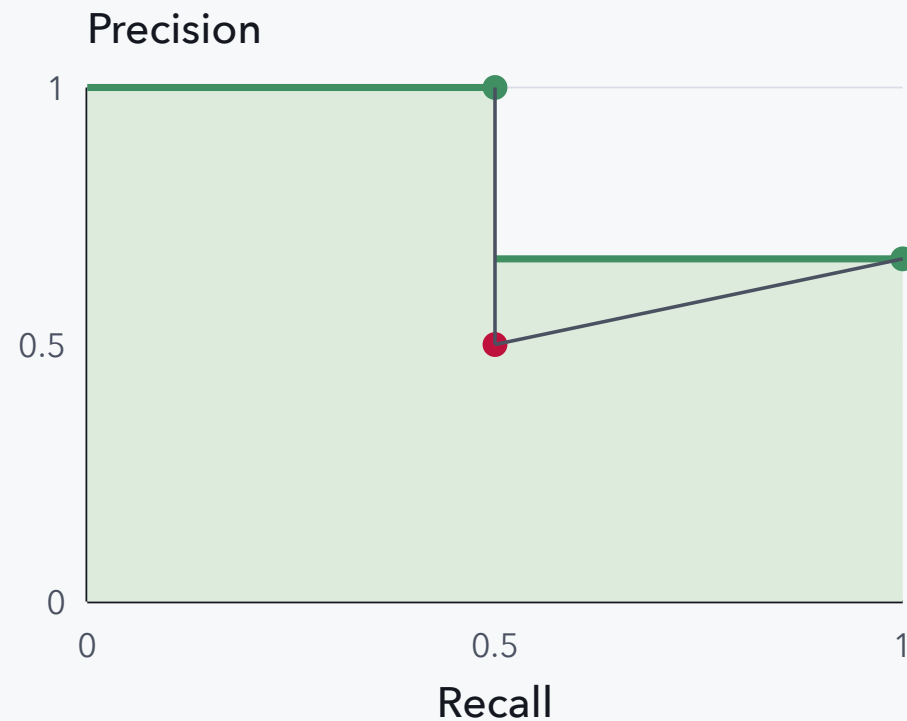


After 3: precision 0.667, recall 1.000.

AP summarizes precision across recall

TWO GROUND-TRUTH OBJECTS

P1	TP	0.95
P2	FP	0.89
P3	TP	0.88



For this toy example, AP is the shaded area under the precision envelope.

Average AP across classes

- dog: AP 0.85
- cat: AP 0.43
- bird: AP 0.76

$$\begin{aligned} \text{mAP} &= \frac{1}{C} \sum_{c=1}^C \text{AP}_c \\ &= \frac{0.85 + 0.43 + 0.76}{3} = 0.68 \end{aligned}$$

The IoU protocol changes what counts as correct

mAP@.50

Match at $\text{IoU} \geq .50$.

Compute AP and average across classes.

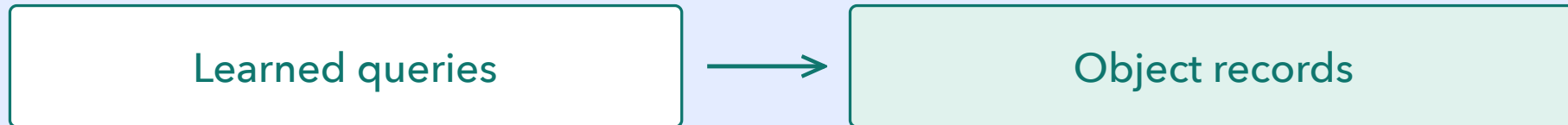
COCO-STYLE AP@.50:.95

Evaluate at .50, .55, ..., .95.

Average across the ten thresholds and categories.

Another answer: set prediction with queries

SECTION 07



Return to the global slots that had no fixed identity.

What if the slots could read the whole image?



slot 1	slot 2	slot 3	slot 4	slot 5
right dog	left dog	empty	empty	empty

The list order is a choice. The objects have not changed.

Earlier: which dog belongs to slot 1?

What if the slots could read the whole image?

$$q_1, q_2, \dots, q_K$$

LEARNED QUERY VECTORS

Each query reads the image features and predicts an object record.

Reuse cross-attention from Beyond Attention

LEARNED QUERIES

Q: query states

Fixed query capacity

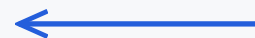
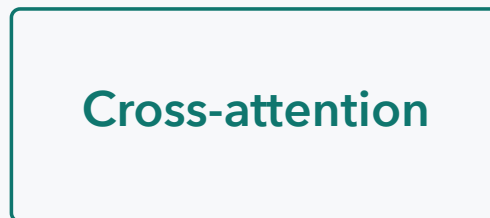
| Query states supply Q .

Reuse cross-attention from Beyond Attention

LEARNED QUERIES

Q: query states

Fixed query capacity



Encoded image features

K, V: keys and values

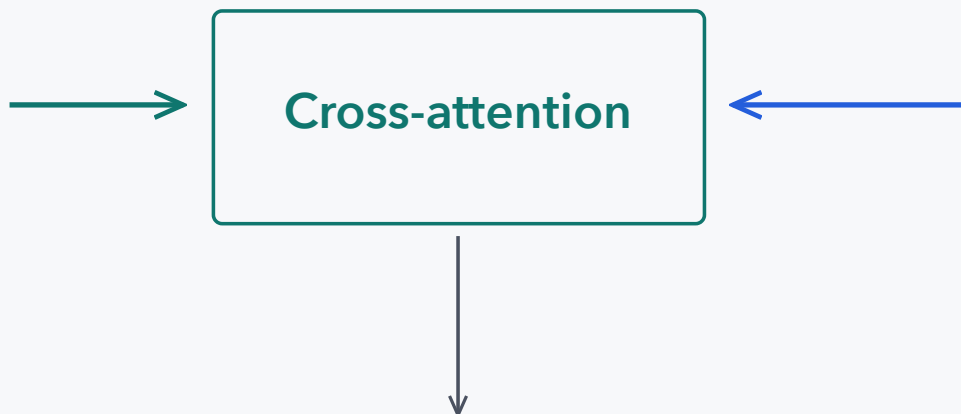
| Encoded image features supply K and V.

Reuse cross-attention from Beyond Attention

LEARNED QUERIES

Q: query states

Fixed query capacity



Encoded image features

K, V: keys and values

One class distribution + one box per query

| Learned queries are another way to index candidate records.

Which prediction should match which target?

THREE QUERY PREDICTIONS

query 1: cat + box

query 2: dog + box

query 3: second dog box

TWO TARGETS

cat

dog

| Choose a one-to-one assignment using class and box cost.

Which prediction should match which target?

THREE QUERY PREDICTIONS

TWO TARGETS

query 1: cat + box



cat

query 2: dog + box



dog

query 3: second dog box

| A low-total-cost assignment pairs each target with one prediction.

Which prediction should match which target?

THREE QUERY PREDICTIONS

TWO TARGETS

query 1: cat + box



cat

query 2: dog + box



dog

query 3: second dog box

no-object

| The unmatched query learns no-object.

Now name the mechanism: bipartite matching

THREE QUERY PREDICTIONS

TWO TARGETS

query 1: cat + box



cat

query 2: dog + box



dog

query 3: second dog box

no-object

Hungarian matching finds a minimum-total-cost one-to-one assignment.

Two solutions to the same output problem

DENSE CANDIDATES

- Spatial locations index records
- Duplicates can remain
- NMS selects among overlapping predictions

DETR-STYLE SET PREDICTION

- Learned queries index records
- One-to-one training discourages duplicates
- Original DETR inference does not require NMS

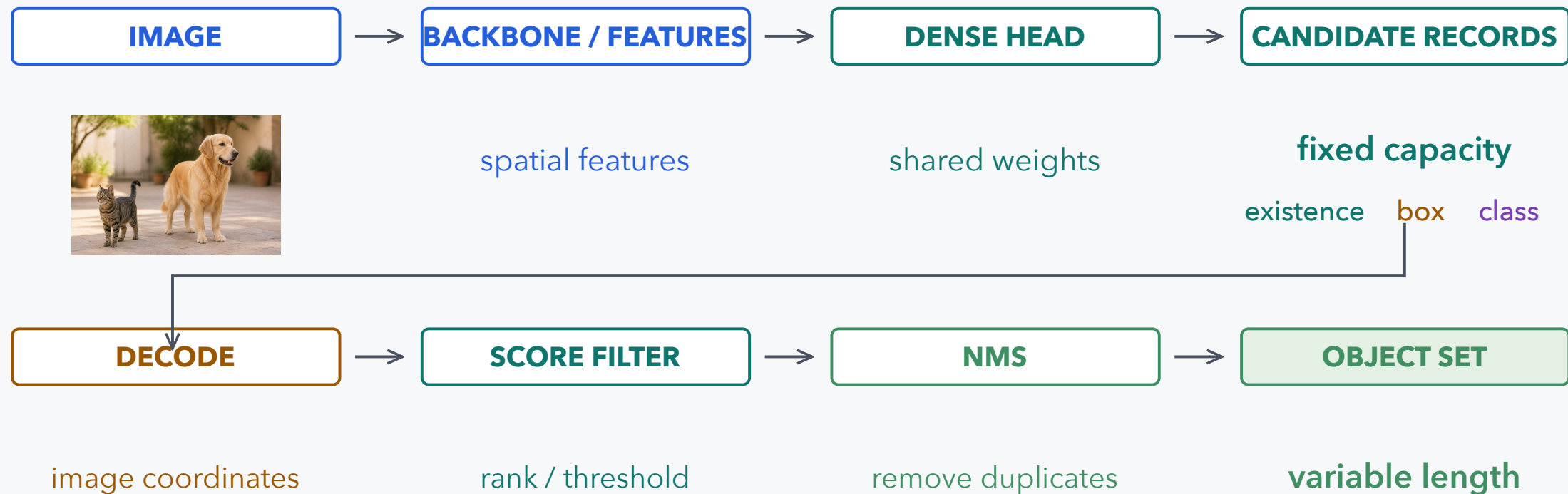
Summary: fixed capacity to a variable set

SECTION 08



Connect the representation, training and inference decisions.

The complete dense-detector path



TRAINING ONLY: assign targets → losses → update the head

Try the same parcel candidates



[Open the offline detector lab](#)

- Move score and IoU thresholds separately
- Step through the greedy NMS choices
- Compare class-aware and class-agnostic grouping

[Download the PyTorch companion notebook](#)

The network can stay fixed-shape. The scene does not have to.

DENSE ANSWER

Image features → spatial candidate records → decode → filter / NMS → returned set

SET-PREDICTION ANSWER

Image features + learned queries → one-to-one-trained candidate set → object records

Three coordinate formats

$$(x_{\min}, y_{\min}, x_{\max}, y_{\max})$$

Corners

$$(x, y, w, h)$$

Top-left and size

$$(x_c, y_c, w, h)$$

Centre and size

Worked coordinate conversion

(130, 330, 240, 310)

Top-left x, y, width, height in pixels.

Image: 1200 × 675 pixels.

CENTRE IN PIXELS

(250, 485)

NORMALIZED CENTRE AND SIZE

(0.208, 0.719, 0.200, 0.459)

Letterboxing transforms the boxes too



$$1200 \times 675 \rightarrow 640 \times 360$$

$$r = 640/1200 = 0.5333$$

$$x' = rx, \quad y' = ry + 140$$

Pad above and below to make a square.

A minimal PyTorch target

```
boxes = torch.tensor([[130., 330., 370., 640.],  
                      [760., 180., 1050., 620.]])  
labels = torch.tensor([2, 1], dtype=torch.int64)  
target = {"boxes": boxes, "labels": labels}
```

Torchvision detection boxes use pixel corners.

Target encoding and decoding are inverses

$$i = 1, \quad j = 2, \quad S = 4$$

$$t_x = 0.48, \quad t_y = 0.64$$

$$x_c = (2 + 0.48)/4 = 0.62$$

$$y_c = (1 + 0.64)/4 = 0.41$$

Why YOLOv1 used $7 \times 7 \times 30$

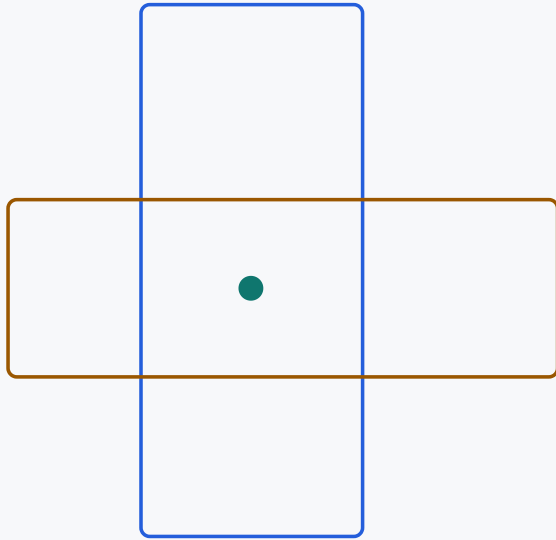
7×7 cells

20 class probabilities per cell

$$20 + 2 \times 5 = 30$$

Each box has x, y, w, h and confidence.

Two predictors do not guarantee two objects

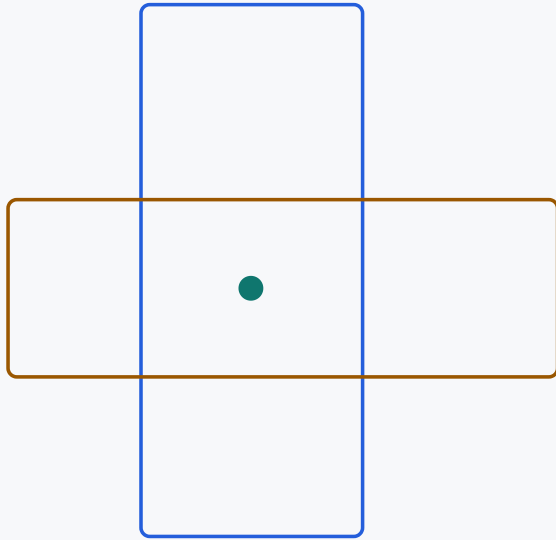


One location; several geometries.

The assignment rule still matters.

| Original YOLOv1 shares one class vector per cell.

Anchors provide reference geometries



One location; several geometries.

The assignment rule still matters.

| Choose a reference shape, then predict offsets from it.

A compact dense prediction head

```
head = nn.Conv2d(channels, anchors * (5 + classes), 1)
raw = head(features)      # [batch, B*(5+C), Hf, Wf]
raw = raw.permute(0, 2, 3, 1)
raw = raw.reshape(batch, Hf, Wf, anchors, 5 + classes)
```

Assignment, target encoding and decoding are separate parts.

Feature pyramids support different object scales



- Fine maps retain spatial detail
- Coarse maps carry broader context
- Several levels contribute candidates

The full binary cross-entropy expression

$$\mathcal{L} = -[y \log p + (1 - y) \log(1 - p)]$$

Positive target with $p = 0.80$

$$\mathcal{L} = 0.223$$

Background target with the same p

$$\mathcal{L} = 1.609$$

Box losses measure geometric error

COORDINATE LOSSES

L1 or Smooth L1 compares coordinates.

OVERLAP LOSSES

IoU-based losses compare regions. GloU also considers the enclosing box.

An additional IoU calculation

$$A = (1, 1, 5, 5)$$

$$B = (3, 2, 7, 6)$$

Continuous corner coordinates.

$$|A| = 16, \quad |B| = 16$$

$$|A \cap B| = 6, \quad |A \cup B| = 26$$

$$\text{IoU} = 6/26 \approx 0.231$$

Class-aware NMS in Torchvision

```
from torchvision.ops import batched_nms

keep = batched_nms(boxes, scores, labels, iou_threshold=0.5)
boxes, scores = boxes[keep], scores[keep]
labels = labels[keep]
```

Run separately for each image. Labels define suppression groups.

Duplicate, wrong-class and low-loU predictions

- A duplicate of an already claimed object is FP
- A wrong-class prediction is FP for that class
- A poorly located box can be FP while its target is FN
- An object with no matched prediction is FN

Toy AP from two recall intervals

$$\text{AP}_{area} = 0.50 \times 1.00 + 0.50 \times \frac{2}{3}$$
$$\approx 0.833$$

Continuous area under the interpolated precision envelope.

Final counts can hide a different ranking

TP → FP → TP

$$AP_{area} \approx 0.833$$

TP → TP → FP

$$AP_{area} = 1$$

A high-level API may already run NMS

```
with torch.inference_mode():  
    pred = model([image_tensor])[0]  
  
    keep = pred["scores"] >= 0.50  
    boxes = pred["boxes"][keep]  
    labels = pred["labels"][keep]
```

Torchvision Faster R-CNN returns post-processed detections.

One-stage and two-stage detector families

ONE STAGE

Predict classes and boxes directly from dense features.

TWO STAGE

Generate proposals, then classify and refine regions.

Sources and further practice

- [YOLO: original dense detector paper](#)
- [DETR: object queries and set prediction](#)
- [Feature Pyramid Networks](#)
- [COCO detection evaluation](#)
- [Offline detector lab](#)
- [PyTorch companion notebook](#)